The

Pulickal Notebook

Volume II

Reconstruction, Algorithms, Operators, Topology, and Experiments

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Abstract

This volume reconstructs the computational, operator-theoretic, topological, information-theoretic, stochastic, and engineering programmes that appeared in *The Pulickal Bible*. It is deliberately separate from the first Pulickal Notebook: its purpose is not to repeat the cleaned recursive-root theory, but to recover the other programmes that were absent from that volume.

The reconstruction begins with the *Pulickal Algorithm*, a proposed multi-directional elimination architecture in which separated computational fronts advance toward a central interface. The original algebraic formula is shown to be incomplete for general dense matrices because cross-coupling blocks cannot be discarded. A corrected formulation is obtained directly from block Gaussian elimination and the Schur complement. This turns the original claim into a precise computational question: whether a particular two-front ordering can reduce critical path length, communication, memory movement, or wall-clock time for a meaningful class of systems.

The second part studies a family of phase-parameterized differential and memory operators. The original interpretation of recursive coefficients as an automatic stability mechanism is rejected. Instead, stability is reduced to explicit spectral and semigroup questions. This separates a syntactically interesting operator from the dynamical claims one might wish to prove about it.

The third part reconstructs the "Cubix" idea as a three-dimensional computational domain rather than as an automatically new tensor algebra. The natural mathematical objects are subdomains, interfaces, separators, local operators, and Schur complements. This permits a rigorous comparison with domain decomposition, multifrontal elimination, and nested-dissection style recursion.

The fourth part develops phase-aware topology. Recursive phase is treated as an additional parameter in a filtration, rather than as a replacement for the boundary operator identity $\partial^2 = 0$. This leads naturally to phase-indexed persistent homology, phase-weighted Hodge operators, and topological signal representations. The machine-learning proposals are then rewritten as explicit feature maps and testable models rather than as claims of automatic noise destruction or guaranteed prediction.

Further chapters treat balanced ternary arithmetic, phase-rotation gates, fault tolerance, complex-amplitude interference, phase-conditioned stochastic systems, thermal topology optimization, lightning precursor detection, financial regime experiments, and possible analytic deformations of the zeta function. Each programme has a stated boundary: where the mathematics is complete, where an external theorem is being used, and where a conjecture or experiment is required.

The goal is a convergent research document in the following sense: each chapter defines its object, states its assumptions, proves what can currently be proved, identifies counterexamples

to stronger statements, and ends at an explicit boundary between established mathematics and future work. No chapter is required to produce a single grand “Pulickal limit.”

How to Read This Volume

Integrity Statement

The historical source for the reconstruction is *The Pulickal Bible*. Its original Cubix, recursive algebra, transform, information, electrodynamic, stochastic, topological, machine-learning, and deep-learning programmes are treated as source ideas. The present volume does not silently preserve claims that the source did not justify.

The logical status words are literal.

- **Theorem / Proposition / Lemma:** proved in the stated model.
- **Definition:** an introduced mathematical object.
- **Conjecture:** a mathematically specified statement whose proof is not supplied.
- **Conjecture Target:** a schema-level open hypothesis whose precise class, baseline, budget, or statistical model is deliberately left to a subsequent research specification.
- **Research Program:** a mathematically specified open investigation.
- **Experiment:** a claim that requires numerical, physical, or empirical implementation.
- **Historical Formulation:** a statement retained to document where a proposed idea came from; it is not thereby endorsed.

A repeated theme is the distinction

$$\text{definition} \not\Rightarrow \text{theorem} \not\Rightarrow \text{engineering performance} \not\Rightarrow \text{physical law}.$$

That distinction is not a limitation of imagination. It is the mechanism by which the programme becomes testable.

Contents

Chapter 1

The Reconstruction Principle

1.1 Why a second notebook is needed

The earlier Pulickal work contained more than one mathematical programme. The recursive-root dynamics can be treated as a self-contained branch-sensitive dynamical system, but the Bible also proposed a computational architecture, recursive operators, topological constructions, hardware primitives, and application-level hypotheses. These objects are logically distinct.

The present volume therefore does not attempt to force every idea into a single algebra. Its organizing principle is instead:

$$\boxed{\begin{array}{l}\text{historical proposal} \to \text{failure analysis} \to \text{minimal repair} \\ \to \text{precise object} \to \text{proof or experiment} \to \text{explicit boundary.}\end{array}}$$

1.2 Convergence without a universal limit

A mathematical document is “convergent” when its questions have well-defined endpoints, not when every construction tends to one number.

For example, a computational algorithm may converge by proving correctness; a topological programme may converge by specifying a stable invariant; an empirical programme may converge by specifying a falsifiable benchmark; and an open theorem may converge by being reduced to a sharply stated conjecture.

Definition 1.1 (Chapter stopping criterion)**.** A chapter is considered mathematically closed at its current level when:

1. all principal objects are defined;
2. all assumptions needed for the principal claims are stated;
3. every claimed theorem in scope has a proof;
4. important failure modes in the same model are addressed;
5. unsupported extensions are explicitly separated as conjectures or experiments.

This criterion is intentionally finite. Asking whether a proof itself can be proved, and then whether its foundations can be proved, is a different foundational project and is not required for an ordinary mathematical research manuscript.

1.3 A status map

The reconstruction can be summarized as follows.

Programme	Status	Correct endpoint
Pulickal two-front elimination	Repairable mathematics	Exact block Schur reduction + performance analysis
Cubix architecture	Research programme	3-D domain/interface decomposition
Recursive differential operators	Repairable mathematics	Operator family + spectral/stability theory
Zeta deformation	Partly settled, partly open	Separate trivial translations from genuinely new deformations
Phase-aware topology	Research programme	Parameterized filtration + persistent homology
Hodge phase operators	Research programme	Weighted/self-adjoint operator analysis
Balanced ternary	Valid engineering programme	Explicit hardware cost model
i-Gate	Valid primitive, novelty open	Phase rotation hardware
Complex amplitudes	Valid mathematical construction	Interference and complex measures
Stochastic phase systems	Research programme	Well-posed SDEs and measurable consequences
Lightning prediction	Experiment	Predictive benchmark with physical sensors
Finance	Experiment	Out-of-sample forecasting and regime detection
Cryptography	Rebuild from scratch	Security proof under a formal adversary model
Universal recursive determinism	Rejected	No general theorem supports it

Chapter 2

Verification Ledger and Second-Pass Audit

2.1 Why an explicit audit record is useful

Notebook II is intended to be extensible. Extensibility creates a particular risk: a later addition can silently change the assumptions under which an earlier theorem was true. The audit record below is therefore part of the mathematical content, not merely editorial bookkeeping.

The finite audit protocol is

$$\begin{gathered}\text{define object} \to \text{state assumptions} \to \text{verify calculations}\\ \to \text{prove claim} \to \text{test stronger claims separately}\\ \to \text{stop at the declared boundary.}\end{gathered}$$

2.2 Correction ledger

Location	Problem detected	Corrected formulation
Two-front counterexample	Residual vector was mis-entered	$Ax_h - b = (0, -2/3, 0)^\top$; the residual norm remains 2/3.
Balanced ternary	Uniqueness ignored arbitrary leading zero digits	Uniqueness is stated for canonical expansions.
Fault tolerance	Minimum of two error probabilities controls neither reliably	Weighted, minimax, or Pareto optimization is used.
Itô system	Cross variation of Brownian drivers was unspecified	An instantaneous cross-covariation density $C(t)$ is introduced; in the jointly Gaussian/Brownian case, independence corresponds to $C = 0$.
Channel capacity	Input constraints and information pattern were implicit	Capacity is defined relative to an admissible input class.
Phase persistence	Two parameters were too easily called a filtration	A poset and inclusion condition are required for multiparameter persistence.
Zeta deformation	Series identity was not clearly separated from analytic continuation	The identity is first stated in $\Re(s + \alpha J) > 1$, then continued meromorphically.
Cryptography	Recursive complexity could be mistaken for hardness	Security is tied to a primitive, threat model, and formal game.
Finance / ECG	Leakage controls were incomplete	Purged temporal or subject-independent evaluation is required where appropriate.

2.3 Independent arithmetic verification

For

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix},$$

the exact solution is

$$x^* = \begin{bmatrix} -3/7 \\ 2/7 \\ 11/7 \end{bmatrix}.$$

The historical two-front output is

$$x_h = \begin{bmatrix} -1/3 \\ 0 \\ 5/3 \end{bmatrix},$$

with

$$Ax_h - b = \begin{bmatrix} 0 \\ -2/3 \\ 0 \end{bmatrix}, \qquad \|Ax_h - b\|_2 = 2/3.$$

For the corrected reduction,

$$B = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad E = [3],$$

so

$$E - DB^{-1}C = 7/3, \qquad b_2 - DB^{-1}b_{\mathrm{out}} = 2/3,$$

and hence $x_2 = 2/7$ and $x_{\mathrm{out}} = (-3/7, 11/7)^\top$. The corrected block reduction therefore agrees exactly with direct solution of the original system.

2.4 Stopping without infinite regress

The audit is complete at a chapter's declared scope once the objects, assumptions, proof status, material counterexamples, and model boundary are explicit. Deeper foundational questions remain legitimate research, but they are not silently counted as defects in an already well-defined theorem.

Part I

The Pulickal Computational Architecture

Chapter 3

The Original Pulickal Algorithm: Historical Form and Failure

3.1 Historical 3-by-3 formulation

The original computational proposal partitions a dense system as

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}, \qquad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \qquad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}, \tag{3.1}$$

where the first and third diagonal blocks are interpreted as the two initial computational fronts. The historical proposal forms

$$S_P = A_{22} - A_{21}A_{11}^{-1}A_{12} - A_{23}A_{33}^{-1}A_{32}. \tag{3.2}$$

It then solves

$$S_P x_2 = \widehat{b}_2$$

and subsequently solves a coupled system for (x_1, x_3).

Historical Formulation

The important historical idea is not the exact displayed formula. It is the *two-front elimination schedule*: begin from separated regions, perform useful local work concurrently, and postpone the central coupling until a reduced problem is available.

3.2 Why the old Schur formula is not general

The blocks A_{13} and A_{31} directly couple the two outer fronts. Therefore the outer variables cannot in general be eliminated independently through A_{11} and A_{33}.

Set

$$B = \begin{bmatrix} A_{11} & A_{13} \\ A_{31} & A_{33} \end{bmatrix}, \qquad C = \begin{bmatrix} A_{12} \\ A_{32} \end{bmatrix}, \qquad D = \begin{bmatrix} A_{21} & A_{23} \end{bmatrix}. \tag{3.3}$$

Then the system is equivalently

$$\begin{bmatrix} B & C \\ D & A_{22} \end{bmatrix} \begin{bmatrix} x_{\text{out}} \\ x_2 \end{bmatrix} = \begin{bmatrix} b_{\text{out}} \\ b_2 \end{bmatrix}, \qquad x_{\text{out}} = \begin{bmatrix} x_1 \\ x_3 \end{bmatrix}. \tag{3.4}$$

Provided B is invertible, exact block elimination gives

$$\boxed{S_{\text{true}} = A_{22} - DB^{-1}C.} \tag{3.5}$$

This is the correct central Schur complement.

Proposition 3.1 (Special case in which the historical formula is exact)**.** *If*

$$A_{13} = 0, \qquad A_{31} = 0,$$

then

$$B = \begin{bmatrix} A_{11} & 0 \\ 0 & A_{33} \end{bmatrix}$$

and

$$S_{\text{true}} = A_{22} - A_{21}A_{11}^{-1}A_{12} - A_{23}A_{33}^{-1}A_{32}.$$

Proof. When $A_{13} = A_{31} = 0$, the inverse of B is block diagonal:

$$B^{-1} = \begin{bmatrix} A_{11}^{-1} & 0 \\ 0 & A_{33}^{-1} \end{bmatrix}.$$

Multiplying $DB^{-1}C$ gives exactly the two historical correction terms. $\square$

3.3 A minimal counterexample

Consider

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}. \tag{3.6}$$

All diagonal blocks are nonzero. The historical central formula gives

$$S_P = 3 - \frac{1}{2} - \frac{1}{2} = 2$$

and

$$\widehat{b}_2 = 2 - \frac{1}{2} - \frac{3}{2} = 0.$$

Thus the historical method gives

$$x_2 = 0.$$

The remaining coupled system is

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix},$$

whose solution is

$$x_1 = -\frac{1}{3}, \qquad x_3 = \frac{5}{3}.$$

But the exact solution of the original system is

$$\boxed{x = \begin{bmatrix} -3/7 \\ 2/7 \\ 11/7 \end{bmatrix}.} \tag{3.7}$$

The residual of the historical output is

$$Ax - b = \begin{bmatrix} 0 \\ -2/3 \\ 0 \end{bmatrix},$$

so

$$\|Ax - b\|_2 = \frac{2}{3} \neq 0.$$

Warning

The historical two-front formula is therefore not a correct general dense-matrix solver. This is an algebraic failure, not a numerical rounding issue.

3.4 The corrected algorithm

Definition 3.2 (Corrected Pulickal two-front reduction)**.** Suppose the outer block

$$B = \begin{bmatrix} A_{11} & A_{13} \\ A_{31} & A_{33} \end{bmatrix}$$

is invertible. The corrected reduction is

$$S_P^{\mathrm{corr}} = A_{22} - DB^{-1}C.$$

First solve

$$S_P^{\mathrm{corr}} x_2 = b_2 - DB^{-1} b_{\mathrm{out}},$$

then recover

$$x_{\mathrm{out}} = B^{-1}(b_{\mathrm{out}} - Cx_2).$$

Theorem 3.3 (Correctness of the corrected reduction)**.** *Under the assumptions above, the corrected two-front reduction returns exactly the unique solution of $Ax = b$ whenever A is nonsingular.*

Proof. The block equations are

$$Bx_{\mathrm{out}} + Cx_2 = b_{\mathrm{out}},$$

$$Dx_{\mathrm{out}} + A_{22}x_2 = b_2.$$

Since B is invertible,

$$x_{\mathrm{out}} = B^{-1}(b_{\mathrm{out}} - Cx_2).$$

Substitution into the second equation yields

$$\left(A_{22} - DB^{-1}C\right) x_2 = b_2 - DB^{-1} b_{\mathrm{out}}.$$

Thus the stated reduction is algebraically equivalent to the original system. □

3.5 What is genuinely new to investigate

The correctness theorem is only the beginning. It does not prove that the two-front schedule is faster.

Research Program 3.4 (Pulickal parallelism question)**.** For a specified matrix or PDE-derived matrix class, determine whether a two-front elimination order can reduce one or more of:

$$\text{critical path}, \quad \text{communication volume}, \quad \text{memory movement},$$
$$\text{synchronization count}, \quad \text{wall-clock time}.$$

The comparison must be performed against an explicitly selected baseline such as standard block LU, nested dissection, or a multifrontal method.

3.6 Critical-path model

Let an algorithm be represented by a directed acyclic graph (DAG) whose vertices are computational tasks and whose edges represent dependencies.

Let

$$D_{\text{crit}} = \max_{\gamma} \sum_{v \in \gamma} w(v)$$

be the weighted critical-path length, where $w(v)$ is the cost of task v.

A two-front schedule is useful only if the dependency graph actually satisfies

$$D_{\text{Pulickal}} < D_{\text{baseline}}$$

under the same task granularity and machine assumptions.

The number of fronts by itself proves nothing about critical-path length.

3.7 Work versus span

A balanced parallel algorithm is constrained by two distinct quantities.

$$W = \sum_{v} w(v) \tag{3.8}$$

is total work, while

$$S = D_{\text{crit}} \tag{3.9}$$

is span.

Idealized parallel time on P processors obeys a lower-bound intuition of the form

$$T_P \gtrsim \max\left(\frac{W}{P}, S\right),$$

while real systems additionally pay communication, synchronization, and memory costs.

Therefore the phrase "halves the computation" must never be used unless a specified machine model and benchmark establish it.

Chapter 4

Generalized Multi-Front Elimination

4.1 Fronts and interfaces

The corrected two-front picture can be generalized. Suppose variables are partitioned into m groups,

$$x = (x^{(1)}, \dots, x^{(m)}, x^{(0)}),$$

where $x^{(0)}$ denotes an interface or core.

After grouping the non-core variables into x_{out}, write

$$\begin{bmatrix} B & C \\ D & E \end{bmatrix} \begin{bmatrix} x_{\text{out}} \\ x_{\text{core}} \end{bmatrix} = \begin{bmatrix} b_{\text{out}} \\ b_{\text{core}} \end{bmatrix}. \tag{4.1}$$

When B is block diagonal or block sparse,

$$B = \operatorname{diag}(B_1, \dots, B_m),$$

local factorizations may proceed concurrently. If B contains limited cross-coupling, one can instead factor a sparse outer interaction graph before forming the core Schur complement.

4.2 Graph formulation

Associate a graph $G = (V, E)$ to a sparse linear system. Eliminating a vertex creates fill between its remaining neighbors. A front is therefore not merely a geometric location; it is a set of variables whose elimination exposes a particular separator.

This yields a mathematically meaningful interpretation:

$$\boxed{\text{Pulickal front} = \text{an elimination region selected for concurrency and locality.}}$$

4.3 Separator viewpoint

Suppose a separator S separates two interior regions L and R. Reordering the variables as (L, S, R) gives

$$A = \begin{bmatrix} A_{LL} & A_{LS} & 0 \\ A_{SL} & A_{SS} & A_{SR} \\ 0 & A_{RS} & A_{RR} \end{bmatrix}.$$

Now A_{LL} and A_{RR} can in principle be eliminated independently:

$$S_S = A_{SS} - A_{SL}A_{LL}^{-1}A_{LS} - A_{SR}A_{RR}^{-1}A_{RS}. \tag{4.2}$$

Here the two-front intuition is exactly correct because the cross-coupling is structurally zero.

This observation is important: the historical formula is not useless. It is a correct special case associated with a genuine graph separation.

Proposition 4.1 (Separated-front exactness)**.** *In the block pattern above, if A_{LL} and A_{RR} are invertible, then the separator Schur complement is exactly the sum of the two independent front contributions.*

Proof. The outer matrix is block diagonal:

$$B = \operatorname{diag}(A_{LL}, A_{RR}).$$

The result follows from the block-Schur calculation of the previous chapter. □

4.4 Research theorem target

Research Program 4.2 (Multi-front separator theorem)**.** Identify matrix graph classes for which a symmetric or near-symmetric collection of fronts yields:

1. independent or weakly coupled local factorizations;
2. bounded separator growth;
3. predictable fill;
4. an asymptotically smaller span than a chosen one-directional ordering;
5. a practical communication advantage on a stated parallel machine model.

This is the point where the Pulickal idea meets established sparse-direct-solver theory rather than competing with it rhetorically.

Chapter 5

The Cubix as a Three-Dimensional Computational Architecture

5.1 The historical Cubix

The source construction places eight computational fronts at the vertices of a three-dimensional cube and directs them toward the geometric center.

$$V = \{(0,0,0),(n,0,0),(0,n,0),(0,0,n),(n,n,0),(n,0,n),(0,n,n),(n,n,n)\}. \tag{5.1}$$

The proposed center is

$$C_P = \left(\frac{n}{2},\frac{n}{2},\frac{n}{2}\right). \tag{5.2}$$

Historical Formulation

The historical intuition is spatial symmetry plus convergent computation. The mathematical repair is to model the physical or computational domain explicitly and to derive the interface problem from the operator being discretized.

5.2 Why a third-order tensor is not automatically a new linear-algebra system

A tensor

$$\mathsf{A} \in \mathbb{R}^{n\times n\times n}$$

is an array of numbers. By itself it does not specify a linear map $\mathbb{R}^n \to \mathbb{R}^n$, nor does it specify a Gaussian-elimination law.

A tensor can encode a multilinear map, a discretized field, a coefficient array, or a data cube.

The associated linear system must be specified separately.

Thus the repaired Cubix abstraction is:

$$\boxed{\begin{array}{l}\text{3-D domain} \to \text{operator} \to \text{discretization} \to \text{graph}\\ \to \text{subdomains} \to \text{interfaces} \to \text{Schur system.}\end{array}}$$

5.3 Eight-subdomain partition

Let a physical domain $\Omega \subset \mathbb{R}^3$ be partitioned into eight subdomains

$$\Omega = \bigcup_{\ell=1}^{8} \Omega_\ell.$$

Each subdomain has local unknowns u_ℓ and shares interface degrees of freedom with neighboring subdomains.

The discrete system may be written schematically as

$$A = \begin{bmatrix} A_{II} & A_{IS} \\ A_{SI} & A_{SS} \end{bmatrix}, \tag{5.3}$$

where I denotes all subdomain-interior variables and S denotes interface variables.

If

$$A_{II} = \mathrm{diag}(A_1, \dots, A_8),$$

then the eight local algebraic factorizations can be performed independently up to the shared interface assembly stage.

The exact interface Schur complement is

$$\boxed{S_S = A_{SS} - A_{SI} A_{II}^{-1} A_{IS}.} \tag{5.4}$$

This is the mathematically rigorous version of a converging multi-front architecture.

5.4 The center is an interface, not a mystical point

The geometric center

$$C_P$$

may be useful as a partition marker, but a single coordinate is not generally the mathematical locus of all interaction.

For PDEs and finite-element discretizations, the actual coupling is carried by an interface set whose dimension depends on the problem. The interface may be a surface, curve, or more general separator in the discretized graph, depending on the geometry and discretization.

Hence the proper statement is:

$$\text{Pulickal Core} = \text{reduced interface problem},$$

not necessarily

$$\text{Pulickal Core} = \text{one physical point}.$$

5.5 Dependency depth

Suppose the spatial domain is recursively partitioned into halves in each coordinate. After L levels,

$$8^L$$

subdomains may be produced, so an ideal balanced recursion can have

$$L = \mathcal{O}(\log n)$$

levels when the characteristic number of subdivisions per coordinate grows exponentially with depth. Equivalently, if the problem has n degrees of resolution per coordinate, then $8^L \asymp n^3$ gives $L = \mathcal{O}(\log n)$.

But this says only that the *partition tree* has logarithmic depth. It does not prove an $\mathcal{O}(\log n)$ solve time.

Warning

A logarithmic recursion depth is not the same thing as logarithmic arithmetic complexity. Matrix factorization still carries its own work and fill costs.

5.6 Cubix research target

Research Program 5.1 (Eight-front Cubix benchmark)**.** Construct a family of three-dimensional sparse systems, for example from an elliptic PDE on a structured mesh. Compare:

standard ordering, nested dissection, multifrontal,

eight-front Pulickal ordering.

Measure:

$$W, \quad S, \quad \text{fill}, \quad \text{memory}, \quad \text{communication}, \quad T_P.$$

A positive result would be meaningful. A negative result would also be meaningful, because it would identify the matrix classes for which geometric symmetry does not translate into computational advantage.

Chapter 6

Sparse Direct Solvers: Fill, Span, and Stability

6.1 Four layers of a solver claim

The corrected Schur formula establishes algebraic correctness, but an algorithmic claim has at least four additional layers:

$$\text{algebraic equivalence,} \qquad \text{numerical stability,}$$
$$\text{work/span complexity,} \qquad \text{communication and memory cost.}$$

Sparse direct methods explicitly separate ordering and symbolic analysis from numerical factorization and triangular solution [9].

6.2 Permutation and fill-reducing orderings

For a permutation matrix P, the reordered matrix is

$$\widetilde{A} = PAP^\top.$$

In an SPD problem one can seek a Cholesky factorization $\widetilde{A} = LL^\top$. For a general nonsymmetric problem, LU with a stated pivoting strategy or QR may be appropriate. The graph ordering and numerical factorization are therefore distinct parts of the algorithm.

Eliminating a vertex in the sparsity graph can create fill edges among its remaining neighbors. Separators reduce this effect by isolating subgraphs whose interiors can be processed before the separator. This is the structural reason nested dissection and multifrontal methods expose parallel subproblems [10, 11].

Proposition 6.1 (Exact parallelism from a separated outer block)**.** *Suppose*

$$A = \begin{bmatrix} A_L & 0 & A_{LS} \\ 0 & A_R & A_{RS} \\ A_{SL} & A_{SR} & A_S \end{bmatrix}$$

with A_L and A_R invertible. Then the separator reduction is

$$S_S = A_S - A_{SL}A_L^{-1}A_{LS} - A_{SR}A_R^{-1}A_{RS},$$

and the two local elimination computations are independent until their contributions are assembled at the separator.

Proof. Group (x_L, x_R) into one outer vector. The outer block is $\operatorname{diag}(A_L, A_R)$, so its inverse is block diagonal. The block Schur formula then yields the displayed expression, and no data dependency couples the two local solves before the separator stage. □

6.3 Work, span, and idealized parallel time

Let W be total computational work and S the weighted length of a longest dependency path. Under the basic task model, any execution on P processors satisfies

$$T_P \geq \max\{W/P, S\}.$$

This is a lower bound, not an execution model. Real systems add communication, memory stalls, synchronization, load imbalance, and accelerator overhead. Consequently a reduction in S does not establish a wall-clock speedup if W or communication grows substantially.

6.4 Arithmetic cost versus memory cost

For a sparse factorization, useful measurements include

flops, nonzeros in factors, peak memory,

bytes transferred, messages, span, wall-clock time.

The nonzero count captures storage and much of the arithmetic structure, whereas bytes and messages capture costs that become dominant on distributed or memory-bound machines.

6.5 Numerical stability

For a computed solution $\widehat{x}$, a useful residual is

$$r = b - A\widehat{x}.$$

One normalized backward-error style quantity is

$$\eta_{\text{bw}} = \frac{\|r\|}{\|A\|\,\|\widehat{x}\| + \|b\|},$$

with the norm declared in advance, whenever the denominator is nonzero. A structurally excellent ordering can still be a poor solver if its numerical factorization is unstable. For nonsymmetric systems, pivoting and scaling policies must therefore be fixed in the benchmark.

6.6 A fair two-front benchmark

The benchmark should predefine the matrix family, size sequence, sparsity pattern, ordering, factorization strategy, pivoting policy, right-hand sides, processor count, machine topology, numerical precision, stopping criterion, and whether symbolic preprocessing is charged to time. The baseline must be a serious established method appropriate to the same class of matrices. A positive result should be replicated over a size range; a negative result should identify the matrix classes and cost components responsible for failure.

6.7 Research target

The strongest defensible computational question is therefore:

> For which sparse matrix families and machine models does a specified multi-front ordering reduce end-to-end cost while preserving a stated numerical-accuracy target?

This is narrow enough to be falsifiable and broad enough to produce meaningful algorithmic results whether the answer is positive or negative.

Part II

Phase-Parameterized Operators

Chapter 7

Recursive Phase as an Operator Parameter

7.1 Historical notation

The source proposes a sequence of phase-dependent coefficients and an operator of the form

$$\mathcal{D}_P f = \sum_{n=0}^{N} c_n D^n f, \qquad D = \frac{\mathrm{d}}{\mathrm{d}t}. \tag{7.1}$$

The historical coefficients were built from recursive imaginary symbols.

The useful mathematical core is the finite differential operator itself.

7.2 Definition of a phase-parameterized operator

Definition 7.1. Let ϕ belong to a parameter space Φ, let $c_0(\phi), \dots, c_N(\phi) \in \mathbb{C}$, and define

$$\mathcal{D}_\phi = \sum_{n=0}^{N} c_n(\phi) D^n. \tag{7.2}$$

Its symbol is

$$p_\phi(\lambda) = \sum_{n=0}^{N} c_n(\phi)\lambda^n. \tag{7.3}$$

For a modal solution $f(t) = e^{\lambda t}$, with ϕ held fixed as a parameter with respect to t,

$$\mathcal{D}_\phi f = p_\phi(\lambda) e^{\lambda t}.$$

Thus the characteristic polynomial is the correct starting point for stability in the constant-parameter model. If the phase is itself time-dependent, $\phi = \phi(t)$, then the coefficients $c_n(\phi(t))$ become time-dependent, so the fixed-parameter symbol $p_\phi(\lambda)$ is no longer the whole stability calculation; one must analyze the resulting time-dependent equation or state-space system.

Thus the characteristic polynomial is the correct starting point for stability in the fixed-parameter case.

7.3 Stability is not automatic

Consider

$$\mathcal{D}_\phi y = Ay. \tag{7.4}$$

If A is finite-dimensional and the operator is reduced to a first-order state-space system (with the highest-order coefficient invertible so that the reduction is well-defined), stability is governed by the eigenvalues of the resulting system matrix.

There is no general implication

$$\text{phase convergence} \Rightarrow \Re\lambda < 0.$$

Indeed, if the leading coefficient or lower-order coefficients vary in a way that moves a root into the right half-plane, instability follows.

Proposition 7.2 (Spectral criterion for finite-dimensional exponential stability)**.** *For a finite-dimensional linear state-space realization*

$$\dot{z} = M_\phi z,$$

with the usual continuous-time notion of exponential stability,

$$\max_{\lambda \in \mathrm{spec}(M_\phi)} \Re\lambda < 0$$

is necessary and sufficient for exponential stability.

Proof. The solution is $z(t) = e^{tM_\phi} z(0)$. Exponential stability requires every eigenmode to decay, hence every eigenvalue must have negative real part. Conversely, for a finite-dimensional matrix, if all eigenvalues have negative real part then the matrix exponential satisfies an estimate $\|e^{tM_\phi}\| \le Me^{-\alpha t}$ for some $M, \alpha > 0$; thus the system is exponentially stable. □

7.4 Memory is a different operator class

A finite sum of derivatives is local in time. A genuine memory term is naturally written as

$$(\mathcal{V}_K f)(t) = \int_0^t K(t-\tau) f(\tau)\, \mathrm{d}\tau. \tag{7.5}$$

A more complete phase-aware model is

$$\boxed{\mathcal{A}_\phi f = \sum_{n=0}^{N} c_n(\phi) D^n f + \int_0^t K_\phi(t-\tau) f(\tau)\, \mathrm{d}\tau.} \tag{7.6}$$

This separates differential memory surrogates from actual history dependence.

7.5 Laplace-domain interpretation

If the relevant transforms exist, then

$$\mathcal{L}\{D^n f\}(s) = s^n F(s) - \text{initial terms},$$

and

$$\mathcal{L}\{\mathcal{V}_K f\}(s) = \widehat{K}(s) F(s).$$

Hence the transfer operator is modified by

$$p_\phi(s) + \widehat{K}_\phi(s),$$

up to initial-condition terms.

This provides a rigorous route to study "recursive damping" without assigning physical meaning to a formal recursive unit.

Research Program 7.3 (Phase-dependent stability)**.** Classify coefficient families $(c_n(\phi))$ and memory kernels K_ϕ for which the associated evolution is exponentially stable, contractive, passive, or dissipative.

Chapter 8

Phase-Weighted Spectral and Hodge Operators

8.1 Discrete Hodge Laplacian

Let

$$\partial_k : C_k \to C_{k-1}$$

be boundary operators with

$$\partial_{k-1}\partial_k = 0.$$

Given inner products, define the adjoint ∂_k^*.

The classical degree-k Hodge Laplacian is

$$L_k = \partial_{k+1}\partial_{k+1}^* + \partial_k^*\partial_k. \tag{8.1}$$

It is self-adjoint and positive semidefinite.

8.2 Phase-weighted version

Fix a common inner product (for example the Euclidean inner product in the finite-dimensional matrix representation). Let $w_j(\phi) \in \mathbb{R}$ and define

$$L_k^\phi = \sum_{j=1}^{J} w_j(\phi) L_{k,j}. \tag{8.2}$$

If every $L_{k,j}$ acts on the same degree-k finite-dimensional cochain space, is self-adjoint with respect to one fixed common inner product, and

$$w_j(\phi) \geq 0,$$

then L_k^{ϕ} is also self-adjoint and positive semidefinite.

Proposition 8.1 (Positive weighted Hodge combination)**.** *If all L_j act on the same finite-dimensional inner-product space, each satisfies $L_j = L_j^*$ and $L_j \succeq 0$, and $w_j \geq 0$, then*

$$L = \sum_j w_j L_j$$

satisfies

$$L = L^*, \qquad L \succeq 0.$$

Proof. Adjoints distribute over finite sums:

$$L^* = \sum_j w_j L_j^* = L.$$

For every vector x,

$$\langle x, Lx \rangle = \sum_j w_j \langle x, L_j x \rangle \geq 0.$$

□

This theorem supplies a rigorous replacement for multiplying a Hodge Laplacian by complex recursive coefficients and then assuming that the resulting operator remains a physically meaningful positive operator.

8.3 Complex coefficients: what changes

If one instead uses complex weights,

$$\widetilde{L} = \sum_j c_j L_j, \qquad c_j \in \mathbb{C},$$

then $\widetilde{L}$ need not be self-adjoint or positive semidefinite.

That does not make it meaningless. It simply moves the problem into non-self-adjoint spectral analysis.

The relevant objects become:

$$\operatorname{spec}(\widetilde{L}), \qquad \text{pseudospectrum}, \qquad \|\exp(-t\widetilde{L})\|,$$

rather than an automatic energy-minimizing interpretation.

Research Program 8.2 (Phase-Hodge theory)**.** Find phase-weight functions $w_j(\phi)$ or complex structures for which:

1. self-adjointness is preserved or deliberately controlled;

2. harmonic spaces remain interpretable;
3. spectral gaps respond predictably to phase;
4. filtering performance improves on specified graph/simplicial signal classes.

Part III

Zeta and Arithmetic Reconstruction

Chapter 9

Zeta Deformations: What the Simple Formula Actually Does

9.1 The historical deformation

The source proposes

$$\zeta_{\mathcal{H}}(s, J) = \sum_{n=1}^{\infty} \frac{1}{n^{s+\alpha J}}, \qquad J \in \mathbb{C}. \tag{9.1}$$

Whenever $\Re(s + \alpha J) > 1$, the defining Dirichlet series converges absolutely and

$$\boxed{\zeta_{\mathcal{H}}(s, J) = \zeta(s + \alpha J).} \tag{9.2}$$

The right-hand side then supplies the meromorphic continuation in s. Thus the simple deformation is literally a translation in the complex argument and does not produce a new analytic function once continuation is taken into account.

9.2 Pole and zero transport

Suppose ρ is a zero of the ordinary zeta function. Then

$$\zeta_{\mathcal{H}}(s, J) = 0$$

exactly when

$$s + \alpha J = \rho.$$

Hence

$$\boxed{s = \rho - \alpha J.} \tag{9.3}$$

Likewise the pole at 1 is transported to

$$s = 1 - \alpha J.$$

Therefore the simple deformation does not by itself create a new intrinsic zero geometry.

Proposition 9.1 (Translation of the simple zeta deformation)**.** *For fixed J and α, the zero set of $s \mapsto \zeta(s + \alpha J)$ is the translate*

$$Z_J = Z_0 - \alpha J$$

of the ordinary zeta zero set Z_0.

Proof. Immediate from the equivalence

$$\zeta(s + \alpha J) = 0 \iff s + \alpha J \in Z_0.$$

□

9.3 A genuinely nontrivial deformation

To obtain new analytic information, one needs a deformation that changes more than the argument by translation.

Possible directions include

$$\zeta_{\alpha,\beta}(s) = \sum_{n\geq 1} \frac{e^{i\beta g(n)}}{n^s}, \tag{9.4}$$

or

$$L(s;\theta) = \sum_{n\geq 1} \frac{a_n(\theta)}{n^s}, \tag{9.5}$$

with explicit conditions on $a_n(\theta)$.

Then one can ask whether the deformed object has:

analytic continuation, functional equation, Euler product,
zero distribution, prime-sensitive coefficients.

These are difficult problems and should be labeled accordingly.

Research Program 9.2 (Nontrivial deformation programme)**.** The retained working hypothesis is that there may exist phase-weighted Dirichlet-type deformations whose analytic invariants contain information not obtainable from a mere translation of ζ. A mathematically sharp version must first specify the coefficient class, convergence domain, continuation mechanism, and the invariant to be compared. A successful construction must identify the new invariant explicitly.

Chapter 10

Prime Resonance as a Statistical Programme

10.1 Separating metaphor from mathematics

The historical language “primes are resonance nodes” is suggestive but incomplete. A resonance hypothesis requires a measurable statistic.

Let $\mathcal{P} = \{2, 3, 5, \ldots\}$ denote the primes. Let $R(p;\theta)$ be a specified phase-sensitive statistic.

The question becomes whether the distribution

$$\{R(p;\theta) : p \leq X\}$$

differs from an explicit null model in a statistically significant and mathematically interpretable way.

10.2 A null-model framework

At minimum, compare with:

1. uniformly sampled integers of the same range;
2. randomized sequences preserving selected low-order arithmetic statistics;
3. control sets based on almost-primes or residue classes.

For an i.i.d. null with a justified common variance, and provided the estimated null variance is positive, the familiar standardized statistic may be written

$$Z_X = \frac{\overline{R}_{\text{prime}} - \overline{R}_{\text{null}}}{\widehat{\sigma}_{\text{null}}/\sqrt{N}}. \tag{10.1}$$

More generally, because the prime and null samples need not have an i.i.d. variance structure, define the contrast $T_X = \overline{R}_{\text{prime}} - \overline{R}_{\text{null}}$ and calibrate it under the declared null by an estimated sampling variance (or, preferably for a finite bespoke null, by the full null resampling distribution):

$$Z_X^{(0)} = \frac{T_X}{\sqrt{\widehat{\mathrm{Var}}_0(T_X)}}. \tag{10.2}$$

A nonzero value is not by itself a theorem about primes; it is evidence relative to the specified null-generating mechanism and its sampling assumptions. If the estimated variance is zero or poorly estimated, a direct Z-normalization is inappropriate and an exact or resampling-based null distribution should be used instead.

Research Program 10.1 (Prime-phase statistics)**.** Construct phase statistics whose mathematical definition is independent of the primality labels used in fitting, then determine whether the statistics correlate with prime indicators beyond known arithmetic predictors.

Warning

No statement in this chapter implies a new proof or reformulation of the Riemann hypothesis. Any such claim would require a precise theorem relating the statistic to the zero set of ζ or an equivalent arithmetic statement.

Part IV

Information, Logic, and Hardware

Chapter 11

Phase-Conditioned Information Theory

11.1 The historical entropy idea

The source replaces a real entropy with a logarithm whose base is a recursive imaginary quantity.

That construction can be algebraically written but does not by itself supply the axioms or operational interpretation of Shannon entropy.

We therefore make phase an ordinary model variable.

11.2 A phase-conditioned channel

Let Φ be a random phase variable. Consider a channel

$$Y = h(\Phi)X + N, \tag{11.1}$$

where N is a noise variable conditionally independent of X given Φ.

The relevant conditional mutual information is

$$I(X;Y \mid \Phi) = H(Y \mid \Phi) - H(Y \mid X, \Phi). \tag{11.2}$$

The word "capacity" is meaningful only after the admissible coding rules, information patterns, and resource constraints have been fixed. Let I_T and I_R denote the information patterns available to transmitter and receiver. For a blocklength-n model, the operational capacity is defined from the supremum of rates for which there exists a sequence of admissible codes with error probability tending to zero, under the declared channel law, state process, side information, and resource constraints. A single-letter conditional-mutual-information expression is then a theorem only for a channel class for which the corresponding coding theorem applies.

For a memoryless state model in a setting where conditioning on Φ is operationally appropriate, let $\mathcal{A}(\mathsf{I}_T)$ denote the admissible single-letter input laws consistent with the transmitter information pattern, for example those satisfying $\mathbb{E}|X|^2 \le P$. Define the associated single-letter benchmark (retaining C_{cond} as an optional shorthand for this non-operationally-general quantity):

$$\boxed{C_{\mathrm{SL}}(\mathcal{A}(\mathsf{I}_T)) := C_{\mathrm{cond,SL}}(\mathcal{A}(\mathsf{I}_T)) = \sup_{P_{X|\Phi}\in\mathcal{A}(\mathsf{I}_T)} I(X;Y \mid \Phi).} \tag{11.3}$$

In the particular memoryless models whose operational capacity theorem yields this optimization, C_{SL} equals capacity. In general it should be treated as a model-dependent information quantity rather than being called capacity without the corresponding coding theorem.

If the transmitter does not observe Φ, then the admissible class must forbid adaptation to Φ; in the simplest memoryless formulation this means $P_{X|\Phi} = P_X$. If the receiver does not observe Φ, the operational quantity is not the same conditional mutual information and the decoder must instead operate from its actual observation sigma-field. State information may be causal or noncausal, and these cases can lead to different capacity formulas [3]. For continuous variables, mutual information is the primary quantity; the entropy difference above is shorthand only when the relevant differential entropies are well-defined and finite. Thus capacity is a property of the channel together with its information pattern, coding rules, and constraints, not of the notation used to write the logarithm.

11.3 Why an imaginary logarithmic base does not increase capacity

A formal expression such as

$$\log_{e^{i\theta}} a = \frac{\ln a}{i\theta}$$

is generally complex-valued and is branch-dependent for the complex logarithm; the display should therefore be read only after fixing a branch and taking $\theta \neq 0$. It is not automatically an information measure.

Information capacity is an operational supremum over coding schemes. A change of notation in the logarithm cannot by itself create extra distinguishable channel outputs.

Proposition 11.1 (Capacity requires an operational channel)**.** *A claimed capacity increase is meaningful only if the underlying admissible coding problem changes in at least one material respect, such as the channel law, input alphabet, side information, bandwidth, power constraint, decoding resources, or error criterion.*

Proof. Capacity is defined relative to the admissible coding/decoding problem. If the channel and all constraints remain identical, replacing the notation used to describe mutual information does not alter the achievable coding rates. □

Research Program 11.2 (Phase-assisted communication)**.** Study whether phase side information, phase modulation, or phase-conditioned coding creates useful performance improvements under fixed bandwidth and power constraints. The baseline must use the same channel resources.

Chapter 12

Information Capacity: Information Patterns and Resource Constraints

12.1 Capacity is not a notation problem

The phase-conditioned communication programme becomes mathematically complete only after separating three questions:

channel law, who observes the phase, which resources are constrained.

This is standard information-theoretic discipline; channel capacity is defined relative to an admissible coding problem rather than to a symbol chosen for an entropy formula.

12.2 A state-dependent channel

Let Φ be a random channel state and let

$$P_{Y|X,\Phi}(\cdot \mid x, \phi)$$

be the conditional channel law. Let I_T and I_R denote the information available to transmitter and receiver. Examples include no phase information, receiver-only phase information, or phase information at both ends. These cases produce different coding problems.

12.3 Gaussian illustration

For the real scalar channel

$$Y = HX + N,$$

with $N \sim N(0, \sigma_N^2)$ and a Gaussian input of variance P, the conditional mutual information is

$$I(X; Y \mid H) = \frac{1}{2} \log \left(1 + \frac{H^2 P}{\sigma_N^2} \right).$$

If H is random, one must state whether the power is fixed for every state or only on average, and whether H is known to the transmitter. The corresponding capacity formula changes with the information pattern. The example demonstrates why an operational statement is necessary before claiming a phase advantage. If the receiver knows H but the transmitter does not, one obtains a different optimization from the case in which both know H. With transmitter side information, state-dependent input policies may be admissible; without it, they are not. This is part of the channel model rather than a notational choice.

12.4 A resource-preserving invariance principle

Proposition 12.1 (Notation cannot create capacity)**.** *If two channel descriptions have identical input/output laws, encoder and decoder classes, side information, resource constraints, and error criterion, then they have the same operational capacity regardless of the notation used for their information quantities.*

Proof. The two descriptions admit exactly the same set of codes and therefore exactly the same set of achievable rate/error pairs. Their capacity, defined as the supremum of the achievable rates, is consequently identical. □

Proposition 12.2 (Resource-accounted phase comparison)**.** *A phase-aware system and its baseline should be compared under a common physical resource model. If phase estimation, synchronization, side-channel signaling, extra bandwidth, or decoding work is required, that resource belongs in the admissible constraint set. An apparent information gain that disappears after these costs are charged is not a net system-level gain.*

Proof. Capacity is defined over the admissible set of codes. Adding resources changes that set and hence changes the problem; relabeling an existing internal variable does not. The comparison is therefore well-defined only when both systems are evaluated under the same declared constraints plus any explicitly credited additional resource. □

12.5 Net gains

A phase-assisted communication claim should include phase-estimation overhead, synchronization, bandwidth, power, latency, hardware complexity, and decoding cost. A gain that disappears when these resources are charged to the complete system is not a net system-level gain. This is the information-theoretic analogue of the solver benchmark principle: the baseline must consume the same resources.

Chapter 13

Balanced Ternary Arithmetic

13.1 Representation

Every integer admits a balanced ternary expansion

$$n = \sum_{j=0}^{L} d_j 3^j, \qquad d_j \in \{-1, 0, 1\}. \tag{13.1}$$

A simple recursive construction uses the congruence class of n modulo 3: choose $d_0 \in \{-1, 0, 1\}$ so that $n - d_0$ is divisible by 3, then recurse on $(n - d_0)/3$.

Proposition 13.1 (Existence and uniqueness of canonical balanced ternary)**.** *Every integer n has a unique* canonical *finite balanced-ternary representation*

$$n = \sum_{j=0}^{L} d_j 3^j, \qquad d_j \in \{-1, 0, 1\},$$

where $n = 0$ is represented by the single digit $d_0 = 0$ (so $L = 0$), or $n \neq 0$ and the highest digit satisfies $d_L \neq 0$. Thus uniqueness is understood after excluding arbitrary leading zero digits.

Proof. Existence follows by the modulo-3 recursion. At each step there is a unique digit $d_0 \in \{-1, 0, 1\}$ congruent to n modulo 3. Replacing n by $(n - d_0)/3$ strictly decreases its absolute value whenever $n \neq 0$, so the recursion terminates.

For uniqueness, suppose two canonical expansions represent the same integer. After removing any common leading zeroes, let m be the least index at which their digits differ. Subtracting and dividing by 3^m gives

$$d_m - e_m + 3q = 0$$

for some integer q. Since $d_m - e_m \in \{-2, -1, 1, 2\}$, it cannot be divisible by 3. This contradiction proves uniqueness of the canonical expansion. The qualification is necessary because otherwise the zero integer admits infinitely many strings differing only by leading zero digits. $\square$

13.2 Radix economy versus hardware economy

The abstract radix-economy observation that 3 is close to e is not a universal hardware theorem.

A physical implementation must account for

$$\text{state separation}, \quad \text{noise margin}, \quad \text{switching energy},$$
$$\text{routing}, \quad \text{conversion overhead}, \quad \text{device variability}.$$

Thus a fair comparison must define a cost functional.

Definition 13.2 (Hardware cost functional)**.** For an implementation I, define

$$C(I) = \lambda_A A(I) + \lambda_E E(I) + \lambda_T T(I) + \lambda_R R(I),$$

where the terms represent area, energy, delay, and a reliability penalty, and the weights are fixed by the application.

The statement "ternary is better" is meaningful only relative to such a model.

13.3 Arithmetic complexity

Balanced ternary can encode both signs without a separate sign convention and may reduce digit weight in some arithmetic operations.

A rigorous evaluation should compare:

$$\text{addition}, \quad \text{multiplication}, \quad \text{constant multiplication}, \quad \text{negation}, \quad \text{conversion}.$$

Research Program 13.3 (Balanced-ternary hardware benchmark)**.** Implement binary and balanced-ternary versions of the same arithmetic kernel in the same physical technology assumptions. Report area, energy per operation, delay, and error rate under identical precision and throughput targets.

Chapter 14

The i-Gate and Phase-Rotation Hardware

14.1 The mathematical primitive

Define

$$G_i = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}. \tag{14.1}$$

Then

$$G_i \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}.$$

Identifying $x + iy$ with (x, y), this is exactly multiplication by i:

$$i(x + iy) = -y + ix.$$

Thus the mathematical operation is standard.

14.2 Four phase states

The set

$$\{1, i, -1, -i\}$$

corresponds to four equally spaced phase states. A hardware implementation must specify how these states are encoded physically.

An LC resonator can carry a phase relation between voltage and current, but the claim that it supplies a lossless universal complex gate requires a device-level model and measurement.

14.3 A realistic gate model

Let the internal state be

$$z = re^{i\phi}.$$

An ideal phase gate acts as

$$(r, \phi) \mapsto (r, \phi + \pi/2).$$

A noisy physical realization might instead have

$$(r, \phi) \mapsto (r + \eta_r, \phi + \pi/2 + \eta_\phi).$$

Define phase error

$$e_\phi = \mathrm{PV}\left(\widehat{\phi} - \phi_{\text{target}}\right).$$

This produces a direct route to fault detection.

Research Program 14.1 (Phase-gate robustness)**.** Determine the minimum phase separation, signal-to-noise ratio, and energy required for reliable four-state phase discrimination in a chosen circuit technology.

Chapter 15

Phase-Redundant and Fault-Tolerant Hardware

15.1 The Hardware DNA intuition

The historical proposal suggests that an error should appear as a phase drift rather than as an undetected discrete bit flip.

The repaired idea is phase-redundant encoding.

Let an intended state be

$$s = (a, \phi).$$

Suppose a code associates a valid state manifold

$$\mathcal{M} \subset \mathbb{R}_+ \times S^1.$$

Equip the state space with an explicit metric, for example

$$d((a,\phi),(b,\psi))^2 = \frac{(a-b)^2}{\sigma_a^2} + \frac{\mathrm{PV}(\phi-\psi)^2}{\sigma_\phi^2},$$

where $\sigma_a, \sigma_\phi > 0$ determine the relative scale of amplitude and phase errors. Define

$$E(s) = \inf_{m \in \mathcal{M}} d(s,m)^2.$$

A correction rule may choose a nearest admissible point when a minimizer exists. If the minimizer is not unique, a deterministic tie rule or set-valued decoder must be specified.

15.2 Detection and correction probabilities

Let

$$P_{\text{miss}} = \mathbb{P}(\text{error not detected}),$$

and

$$P_{\text{false}} = \mathbb{P}(\text{clean state incorrectly corrected}).$$

A useful code is one for which both are controlled under an explicit noise model.

The two error probabilities are different failure modes and should not be combined through a minimum. The objective $\min\{P_{\text{miss}}, P_{\text{false}}\}$ would reward a design that makes only one error mode small. A meaningful scalar objective is, for fixed nonnegative weights,

$$\min\left(\lambda_1 P_{\text{miss}} + \lambda_2 P_{\text{false}}\right),$$

subject to area, energy, latency, and information-rate constraints. A symmetric alternative is the minimax objective

$$\min \max\{P_{\text{miss}}, P_{\text{false}}\}.$$

If no defensible scalar trade-off is available, the Pareto frontier of achievable pairs $(P_{\text{miss}}, P_{\text{false}})$ is the cleanest mathematical object.

Research Program 15.1 (Phase-redundant fault tolerance)**.** Construct phase/amplitude codes with provable separation under a specified stochastic disturbance model and compare them with binary error-detecting and error-correcting codes at equal energy and bandwidth.

No automatic self-healing property is assumed.

Part V

Complex Amplitudes and Stochastic Systems

Chapter 16

Complex Amplitudes and Interference

16.1 Complex measures

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let

$$\psi \in L^1(\Omega, \mu)$$

be complex-valued.

Define

$$\nu(A) = \int_A \psi \, \mathrm{d}\mu. \tag{16.1}$$

Then ν is a complex measure absolutely continuous with respect to μ.

Define

$$Q(A) = |\nu(A)|^2. \tag{16.2}$$

Then $Q(A) \geq 0$ but need not be additive.

Proposition 16.1 (Interference cross term)**.** *If*

$$\psi = \alpha\psi_1 + \beta\psi_2,$$

then

$$Q(A) = |\alpha|^2 Q_1(A) + |\beta|^2 Q_2(A) + 2\Re\left(\alpha\overline{\beta}\,\nu_1(A)\overline{\nu_2(A)}\right).$$

Proof. Expand

$$|\alpha\nu_1(A) + \beta\nu_2(A)|^2$$

and use $|z|^2 = z\overline{z}$. □

The cross term is the mathematical mechanism behind interference.

16.2 Classical probabilities as a separate object

A classical probability measure must satisfy

$$P(\Omega) = 1, \qquad P(A) \geq 0, \qquad P\left(\bigcup_j A_j\right) = \sum_j P(A_j)$$

for pairwise disjoint measurable sets.

The amplitude-squared set function Q does not generally satisfy additivity.

Thus the proper hierarchy is:

$$\text{complex amplitude} \rightarrow \text{complex measure} \rightarrow \text{quadratic set function.}$$

This is useful mathematics without requiring the object to be called a classical probability measure.

Chapter 17

Stochastic Phase Dynamics

17.1 Phase-modulated stochastic differential equations

Let $X_t \in \mathbb{R}^d$ and $\Phi_t \in \mathbb{R}^m$. A basic model is

$$\mathrm{d}X_t = b(X_t, \Phi_t, t)\,\mathrm{d}t + \sigma(X_t, \Phi_t, t)\,\mathrm{d}W_t, \tag{17.1}$$

$$\mathrm{d}\Phi_t = a(\Phi_t, t)\,\mathrm{d}t + G(\Phi_t, t)\,\mathrm{d}B_t. \tag{17.2}$$

Here $W_t \in \mathbb{R}^q$ and $B_t \in \mathbb{R}^r$ are continuous martingale drivers with Brownian-type quadratic variations; when they are jointly Brownian with a constant cross-covariance, the same formulas apply with constant C. To allow a specified time-varying instantaneous cross-covariation without implying a jointly Brownian vector with constant covariance, specify their quadratic covariation by

$$d\langle W_i, B_j\rangle_t = C_{ij}(t)\,dt,$$

where $C(t)$ is the instantaneous cross-covariation density. The instantaneous joint covariance matrix must be positive semidefinite. In the jointly Gaussian/Brownian setting, $C(t) = 0$ gives independence; for general continuous martingale drivers, zero cross-variation alone does not imply probabilistic independence. The phase variable can influence drift and diffusion without changing the underlying Itô calculus.

17.2 The correct Itô formula

For a sufficiently smooth scalar function $f(x, \phi, t)$, the mixed term is controlled by quadratic covariation. Write

$$d\langle W_i, B_j\rangle_t = C_{ij}(t)\,dt, \qquad C(t) \in \mathbb{R}^{q\times r},$$

so that

$$d\langle X, \Phi\rangle_t = \sigma C(t) G^\top\,dt.$$

With the Frobenius inner product $\langle A, B\rangle_F = \operatorname{tr}(A^\top B)$, Itô's formula is

$$\begin{aligned} \mathrm{d}f = {} & f_t\,\mathrm{d}t + \nabla_x f^\top\,\mathrm{d}X_t + \nabla_\phi f^\top\,\mathrm{d}\Phi_t && (17.3)\\ & + \frac{1}{2}\operatorname{tr}\left(\sigma\sigma^\top\nabla^2_{xx} f\right)\,\mathrm{d}t + \frac{1}{2}\operatorname{tr}\left(GG^\top\nabla^2_{\phi\phi} f\right)\,\mathrm{d}t && (17.4)\\ & + \left\langle \sigma C(t) G^\top, \nabla^2_{x\phi} f\right\rangle_F\, dt. && (17.5) \end{aligned}$$

If W and B are independent, $C(t) = 0$ and the mixed term vanishes. When the individual drivers are standard Brownian components with the declared cross-covariation density, admissible instantaneous cross-covariances satisfy

$$\begin{bmatrix} I_q & C(t) \\ C(t)^\top & I_r \end{bmatrix} \succeq 0,$$

which is the instantaneous covariance compatibility condition for those Brownian marginals. Thus the formula is determined by the stochastic covariance structure rather than by notation.

17.3 Interference kernels

For a real stochastic phase field $\phi_t(\omega)$ define

$$K_t(\omega, \omega') = \mathbb{E}\left[e^{i(\phi_t(\omega) - \phi_t(\omega'))}\right]. \tag{17.6}$$

If

$$K_t(\omega, \omega') \to 0$$

for $\omega \neq \omega'$, cross-interference may decay under suitable integrability conditions.

Research Program 17.1 (Phase-kernel decoherence)**.** Find sufficient conditions on the phase process under which

$$K_t(\omega, \omega') \to 0$$

and quantify the rate of decay. Compare independent noise, correlated noise, Ornstein–Uhlenbeck phases, and common-mode phase noise.

This is a mathematically focused replacement for the broad statement that "random noise annihilates."

Chapter 18

Stochastic Phase Systems: Well-Posedness and Observable Consequences

18.1 What makes an SDE a model

A stochastic phase model requires a state space, an initial condition or law, measurable coefficients, a filtration, and a covariance structure. For

$$dZ_t = b(Z_t, t)\,dt + \Sigma(Z_t, t)\,dW_t,$$

a theorem on existence and uniqueness applies only after the relevant regularity and growth conditions are supplied. The recursive appearance of a coefficient is not a substitute for these hypotheses.

18.2 Positive semidefinite diffusion covariance

For a real diffusion matrix Σ, the instantaneous covariance is

$$a(Z_t, t) = \Sigma(Z_t, t)\Sigma(Z_t, t)^\top \succeq 0.$$

Thus an arbitrary complex expression cannot simply be inserted in a variance slot. One may work with complex state variables by identifying $\mathbb{C}^n$ with $\mathbb{R}^{2n}$, but the covariance of the resulting real process must still be positive semidefinite.

18.3 Gaussian phase differences

If a phase difference

$$\Delta\phi_t = \phi_t(\omega) - \phi_t(\omega')$$

is centered Gaussian with variance V_t, then its coherence factor is

$$\mathbb{E}e^{i\Delta\phi_t} = e^{-V_t/2}.$$

Hence the decisive quantity for interference suppression is the variance of the difference, not the variance of either phase alone. Strong common-mode noise may leave V_t small and therefore preserve coherence.

18.4 Finite interference test

For amplitudes $a_1, \ldots, a_m \in \mathbb{C}$, set

$$A_t = \sum_{j=1}^{m} a_j e^{i\phi_j(t)}.$$

Then

$$\mathbb{E}|A_t|^2 = \sum_j |a_j|^2 + \sum_{j \neq k} a_j \overline{a_k} \mathbb{E}e^{i(\phi_j - \phi_k)}.$$

This identity gives a finite and directly measurable criterion for whether a proposed phase process suppresses cross-interference.

Part VI

Electrodynamics and Thermal Reconstruction

Chapter 19

Memory-Dependent Electrodynamics

19.1 The constitutive-law viewpoint

The historical Hyper-AC proposal modifies Ohm's and Faraday's laws by inserting recursive complex terms.

A more controlled route starts from Maxwell's equations and modifies the constitutive relation.

For example,

$$\mathbf{D}(t) = \epsilon_\infty \mathbf{E}(t) + \int_{-\infty}^{t} K_E(t-\tau)\mathbf{E}(\tau)\,\mathrm{d}\tau. \tag{19.1}$$

Similarly, a magnetic-memory law may be written

$$\mathbf{B}(t) = \mu_\infty \mathbf{H}(t) + \int_{-\infty}^{t} K_H(t-\tau)\mathbf{H}(\tau)\,\mathrm{d}\tau. \tag{19.2}$$

These convolutions are understood only when the stated integrals exist; for example, one may assume a causal kernel in $L^1(0,\infty)$ together with a compatible field class. The kernels may depend on a phase parameter:

$$K_E = K_E(\tau;\phi), \qquad K_H = K_H(\tau;\phi).$$

19.2 Energy and passivity

A physically meaningful constitutive law should be checked for passivity. A simple frequency-domain diagnostic is whether the relevant impedance or susceptibility has a nonnegative dissipative component in the appropriate convention.

One must never infer energy absorption from notation.

Research Program 19.1 (Phase-dependent constitutive media)**.** Determine whether a physically realizable kernel family $K(\tau;\phi)$ can produce a useful transient response without violating passivity, causality, stability, or material realizability.

19.3 Recursive impedance as a transfer operator

Instead of writing

$$Z_{\mathcal{H}} = \sum_n R_n i_n$$

and assigning it unexplained physical meaning, define an operator-valued impedance

$$V(s) = Z(s;\phi)I(s). \tag{19.3}$$

Then measure or calculate

$$Z(\omega;\phi) = R(\omega;\phi) + iX(\omega;\phi).$$

Now resonance shifts, damping, and bandwidth changes are measurable quantities.

Chapter 20

Thermal Topology and Dark-Silicon Optimization

20.1 Classical thermal model

Let $T(x)$ denote steady temperature and $k(x)$ thermal conductivity, with $k(x) > 0$ on the modeled material region. A basic steady-state model is

$$-\nabla \cdot (k(x)\nabla T) = q(x) \qquad \text{in } \Omega, \tag{20.1}$$

with suitable boundary conditions.

For the transient model, assume $\rho > 0$ and $c_p > 0$ in the modeled material. Then

$$\rho c_p \frac{\partial T}{\partial t} - \nabla \cdot (k\nabla T) = q. \tag{20.2}$$

20.2 Topology optimization

Let Ω_{void} denote intentional cooling voids or channels.

A generic objective is

$$\min_{\Omega_{\text{void}}} \left[\max_{x\in\Omega} T(x) + \lambda\, C(\Omega_{\text{void}}) \right] \tag{20.3}$$

subject to manufacturing and mechanical constraints.

20.3 Topological descriptors

A thermal field can be converted into superlevel or sublevel sets:

$$X_\tau = \{x : T(x) \geq \tau\}. \tag{20.4}$$

Persistent homology of X_τ can describe the appearance and disappearance of connected hot regions, channels, and voids.

The repaired idea is therefore:

$$\boxed{\text{thermal PDE} + \text{geometry} + \text{topological descriptors} + \text{optimization.}}$$

Research Program 20.1 (Thermal-bottleneck topology)**.** Test whether persistent topological features of temperature or heat-flux fields predict thermal bottlenecks earlier or more reliably than standard scalar summaries such as maximum temperature, average temperature, and gradient norms.

Warning

A persistent hole in a topological representation does not, by itself, prove that the corresponding physical region should be perforated. The causal engineering step requires the PDE and optimization model.

Part VII

Lightning, Signal Processing, and Machine Learning

Chapter 21

Lightning Precursor Detection

21.1 The hypothesis

The historical Lightning Harvester proposes that measurable electrical or ionization precursors might provide advance warning of a strike. Existing measurements show that initial electric-field changes and VHF activity occur during lightning initiation, making the sensing premise physically meaningful; prediction remains a separate statistical question [4, 5, 6].

The defensible question is:

$$\boxed{\text{Can measurable pre-strike features improve prediction over a baseline?}}$$

21.2 Sensor vector

Let

$$Y_t = \begin{bmatrix} E_t \\ B_t \\ VHF_t \\ O_t \\ M_t \end{bmatrix} \tag{21.1}$$

represent, for example, local electric field, magnetic measurements, radio frequency/VHF activity, optical activity, and meteorological variables.

Let

$$Z_{t,\Delta} = \mathbf{1}\{\text{lightning initiation within } \Delta\}.$$

We seek

$$\mathbb{P}(Z_{t,\Delta} = 1 \mid Y_{[t-L,t]}).$$

21.3 Evaluation

A proper benchmark must use:

precision, recall, false-alarm rate,
lead time, calibration, event-based rather than sample-count accuracy.

Because initiation events are sparse and temporally clustered, high sample-wise accuracy can coexist with poor operational prediction. The benchmark should define the warning horizon, duplicate-alert rule, event-credit rule, and minimum lead time before an alert is counted as successful.

A claim such as "nanosecond prediction" is meaningful only when the timestamping, sensor bandwidth, and physical definition of event time are explicitly stated.

Research Program 21.1 (Lightning phase benchmark)**.** Compare standard time-frequency and electrical precursor features against a phase-aware feature model. Use strict temporal train/test separation and report lead-time distributions, not only a single accuracy number.

No prediction probability is assumed to be 1.

Chapter 22

Phase-Aware Persistent Topology

22.1 The repaired filtration

Let X be a point cloud and let $\epsilon > 0$. Let ϕ be a phase parameter, and define

$$\eta(\phi) > 0. \tag{22.1}$$

A phase-aware Vietoris–Rips complex may be defined by

$$K(X, \epsilon, \phi) = \{\sigma \subseteq X : \operatorname{diam}(\sigma) \leq \epsilon\, \eta(\phi)\}\,. \tag{22.2}$$

The recursive phase is now a parameter rather than a supposed source of an automatic limit.

22.2 The original cosine modifier

A particularly simple choice inspired by the historical construction is

$$\eta(\phi) = \cos\left(\frac{\pi - \phi}{2}\right) \tag{22.3}$$

on a domain where this quantity remains strictly positive, as required by the repaired filtration above.

One can instead enforce positivity by

$$\eta(\phi) = \delta + \left|\cos\left(\frac{\pi - \phi}{2}\right)\right|, \qquad \delta > 0. \tag{22.4}$$

The latter is globally positive but changes the exact historical form.

22.3 Phase as a second filtration parameter

For $\epsilon_1 \leq \epsilon_2$ and fixed ϕ we recover ordinary filtration monotonicity:

$$K(X, \epsilon_1, \phi) \subseteq K(X, \epsilon_2, \phi).$$

If $\eta(\phi)$ itself varies with ϕ, then the pair (ϵ, ϕ) defines a parameterized family. To call it a genuine multiparameter filtration, introduce a poset P on the parameter pairs and require

$$(\epsilon, \phi) \preceq (\epsilon', \phi') \implies K(X, \epsilon, \phi) \subseteq K(X, \epsilon', \phi').$$

For a product-order model this means that the model must be monotone in both parameter coordinates. Without such an order-preserving inclusion rule, the family is a parameterized collection of complexes rather than a persistence module.

The resulting persistence object is not a single barcode in general. In multiparameter persistence the algebraic object is a functor from the chosen parameter poset to vector spaces; there is generally no complete one-dimensional-style barcode classification [13, 7, 8].

Proposition 22.1 (When phase is only a reparameterization)**.** *Suppose $\eta(\phi) > 0$ and define $r = \epsilon\eta(\phi)$. If*

$$K(X, \epsilon, \phi) = K_0(X, r)$$

for a one-parameter family K_0, then the phase-aware family introduces no new combinatorial filtration values beyond those already indexed by r. In this case phase is a reparameterization of scale rather than an additional topological degree of freedom.

Proof. If two parameter pairs (ϵ_1, ϕ_1) and (ϵ_2, ϕ_2) produce the same effective scale r, they produce the same complex. Thus the family factors through the single scalar map $(\epsilon, \phi) \mapsto r$, so its distinct complexes are already contained in the one-parameter family $K_0(X, r)$. □

A nontrivial phase construction should therefore alter the combinatorial comparison itself, not only its overall scale. One option is a phase-dependent dissimilarity

$$d_{ij}(\phi) = g_\phi(d(x_i, x_j))$$

with an edge rule $d_{ij}(\phi) \leq \epsilon$. Another is an anisotropic metric, provided the metric matrix is positive definite:

$$d_{M_\phi}(x_i, x_j) = \sqrt{(x_i - x_j)^\top M_\phi (x_i - x_j)}, \qquad M_\phi \succ 0.$$

If only $M_\phi \succeq 0$ is assumed, the resulting object is generally a pseudometric rather than a metric. Both versions can define useful adjacency constructions, but the distinction matters when invoking metric-specific results. These models can change edge ordering and therefore the sequence of topological events. This is a stronger candidate for additional information than a scalar rescaling of ϵ, but it also introduces an additional modeling burden: positivity, symmetry, parameter regularity, and stability must be checked for M_ϕ or g_ϕ.

Research Program 22.2 (Phase-persistence)**.** Study whether a two-parameter persistence invariant

$$\mathcal{P}(X; \epsilon, \phi)$$

contains information that cannot be recovered from ordinary one-parameter persistence at a fixed phase.

Chapter 23

Topological Signal Processing

23.1 A simplicial signal model

Let C_k denote oriented k-chains and C_k^* the corresponding cochain space. A degree-k signal is represented in coordinates by a cochain vector

$$f_k \in C_k^*.$$

The Hodge Laplacian

$$L_k = B_k^\top B_k + B_{k+1} B_{k+1}^\top$$

provides a positive semidefinite operator under the standard Euclidean inner product on cochain coordinates [26].

A phase-dependent filter can be defined spectrally by

$$H_\phi(L_k) = \sum_{j=0}^{J} a_j(\phi) L_k^j. \tag{23.1}$$

Then

$$\widetilde{f}_k = H_\phi(L_k) f_k.$$

23.2 Persistence thresholding as a separate stage

Let $\mathcal{D}(f)$ denote the persistence diagram associated with a selected filtration of the signal. Define a feature score

$$s(b, d) = d - b.$$

For threshold τ retain

$$\mathcal{D}_\tau(f) = \{(b,d) \in \mathcal{D}(f) : d - b \geq \tau\}.$$

This gives a mathematically explicit topological denoising stage.

It does not imply that every transient is noise.

Research Program 23.1 (Pulickal topological filter)**.** Compare:

spectral filter, wavelet filter, Kalman-style estimator,
ordinary persistence thresholding, phase-aware persistence filtering.

The comparison should use identical reconstruction and error criteria.

Chapter 24

Phase-Aware Topological Machine Learning

24.1 Feature pipeline

A rigorous pipeline is

$$X \longrightarrow \mathcal{D}_\phi(X) \longrightarrow \Psi_\phi(\mathcal{D}_\phi(X)) \longrightarrow \text{classifier}. \tag{24.1}$$

Here Ψ_ϕ is a vectorization map.

24.2 A repaired persistence image

Let (b, p) be birth-persistence coordinates. Let

$$G_\sigma(u, v; b, p) = \exp\left(-\frac{(u-b)^2 + (v-p)^2}{2\sigma^2}\right).$$

A phase-aware persistence image can be defined by

$$\boxed{I_\phi(u, v) = \sum_{(b,p)\in\mathcal{D}_\phi(X)} \rho_\phi(b, p) G_{\sigma(\phi)}(u, v; b, p),} \tag{24.2}$$

with

$$\rho_\phi(b, p) \geq 0, \qquad \sigma(\phi) > 0.$$

This is a standard type of vectorization with a phase-dependent parameter [12].

24.3 Recursive averaging

If a sequence of phases ϕ_n is used, define

$$I_N(u,v) = \sum_{n=1}^{N} w_n I_{\phi_n}(u,v), \qquad w_n \geq 0, \quad \sum_{n=1}^{N} w_n = 1. \tag{24.3}$$

Unlike the historical formula, this does not require summing complex symbols in the output representation.

24.4 Kernel construction

Given a real phase-indexed feature map

$$\widetilde{\Phi}(X,\phi) := \Phi_\phi(X) \in \mathbb{R}^m,$$

(for a fixed common phase this reduces to the original notation), define

$$K((X,\phi_X),(Y,\phi_Y)) = \exp\left(-\gamma\|\widetilde{\Phi}(X,\phi_X) - \widetilde{\Phi}(Y,\phi_Y)\|_2^2\right). \tag{24.4}$$

For $\gamma > 0$, this is a positive semidefinite Gaussian kernel.

If a more exotic distance is proposed, positive semidefiniteness must be proved.

Proposition 24.1 (Gaussian kernel on Euclidean feature space)**.** *For $\gamma > 0$, the kernel*

$$K(x,y) = e^{-\gamma\|x-y\|_2^2}$$

is positive semidefinite.

Proof. For any finite points $x_1, \ldots, x_N \in \mathbb{R}^m$ and coefficients $c_1, \ldots, c_N \in \mathbb{C}$, the Gaussian admits the Fourier representation

$$e^{-\gamma\|x-y\|_2^2} = c_{m,\gamma} \int_{\mathbb{R}^m} e^{-\|\xi\|_2^2/(4\gamma)} e^{i\xi\cdot(x-y)}\, d\xi,$$

where $c_{m,\gamma} > 0$ is the normalizing constant. Hence

$$\sum_{i,j=1}^{N} c_i \overline{c_j} K(x_i,x_j) = c_{m,\gamma} \int_{\mathbb{R}^m} e^{-\|\xi\|_2^2/(4\gamma)} \left|\sum_{i=1}^{N} c_i e^{i\xi\cdot x_i}\right|^2 d\xi \geq 0.$$

Thus every finite Gram matrix is positive semidefinite. □

This standard result is a safe baseline: any proposed Pulickal kernel can be compared to it rather than assuming that a complicated recursive coordinate system automatically produces a valid SVM kernel.

Research Program 24.2 (RI-TML benchmark)**.** Compare ordinary persistence images against phase-aware persistence images under identical classifiers. Evaluate:

$$\begin{gathered}\text{test accuracy}, \quad \text{AUROC}, \quad \text{calibration}, \\ \text{robustness to noise}, \quad \text{feature dimension}, \quad \text{inference cost}.\end{gathered}$$

Chapter 25

Topological Deep Learning: A Reconstructed Programme

25.1 Simplicial layers

Let a network state contain signals on vertices, edges, and higher-dimensional simplices. Using cochain coordinates consistently,

$$h_\ell^{(k)} \in C_k^*.$$

A generic phase-aware layer is

$$h_{\ell+1}^{(k)} = \sigma\left(W_\ell^{(k)} h_\ell^{(k)} + \sum_{j\in\mathcal{N}_k} A_{kj}(\phi_\ell) h_\ell^{(j)} + b_\ell^{(k)}\right). \tag{25.1}$$

The matrices $A_{kj}(\phi_\ell)$ should be understood as linear maps $C_j^* \to C_k^*$ (represented by conformable matrices after choosing bases); they can be built from incidence and Hodge operators.

25.2 A topological regularizer

Let $\mathcal{D}_{\text{data}}$ be a persistence diagram of the input and $\mathcal{D}_{\text{latent}}$ one from a latent representation. A topological regularizer can be written abstractly as

$$L_{\text{top}} = d_{\text{TDA}}(\mathcal{D}_{\text{data}}, \mathcal{D}_{\text{latent}}), \tag{25.2}$$

where d_{TDA} is a chosen stable metric or kernel-based discrepancy.

The total loss becomes

$$\boxed{L = L_{\text{task}} + \lambda_{\text{top}} L_{\text{top}}.} \tag{25.3}$$

This retains the historical intuition that topology should constrain learning, but keeps ordinary optimization and differentiability requirements visible.

25.3 Vanishing gradients

The statement that a phase limit automatically eliminates the vanishing-gradient problem is not mathematically valid.

A valid study must calculate

$$\left\| \frac{\partial h_L}{\partial h_0} \right\|$$

and determine whether its expected or worst-case norm collapses.

Research Program 25.1 (Phase-aware gradient stability)**.** Identify architectures and parameter ranges for which phase-dependent simplicial operators improve gradient propagation without increasing instability elsewhere.

Chapter 26

Persistent Topology: Stability, Nonredundancy, and Learning Hygiene

26.1 Stability as a mathematical baseline

Persistent homology has a stability theory: under the standard hypotheses of the stability theorem, perturbing a tame filtering function in the sup norm perturbs its persistence diagram by at most the same scale in bottleneck distance,

$$d_B(\mathrm{Dgm}(f), \mathrm{Dgm}(g)) \leq \|f - g\|_\infty.$$

This is a reason to use persistence in noisy data analysis, but it is not a predictive-performance theorem. Stability controls sensitivity of the representation; it does not establish that a particular representation improves classification, forecasting, or control [1].

26.2 Persistence images

A persistence image maps a persistence diagram to a finite-dimensional vector by smoothing weighted birth-persistence points. The construction admits stability guarantees and can be combined with ordinary vector-based classifiers [12]. Notebook II therefore treats persistence images as a baseline representation rather than assuming that a phase-modified version must be better.

26.3 Detecting trivial phase redundancy

Suppose

$$K(X, \epsilon, \phi) = K(X, \epsilon\eta(\phi)).$$

Then the pair (ϵ, ϕ) may contain only one effective scale. As a topological filtration parameter, phase is structurally informative only when its effect cannot be absorbed into such a scalar change (unless phase is also supplied as a separate downstream feature). Possible nontrivial mechanisms include phase-dependent edge weights, anisotropic distances, local metrics, or coupled filtrations.

26.4 A statistical nonredundancy test

Let $R_0(X)$ be an ordinary representation and $R_\phi(X)$ a phase-aware one. Define a predeclared test loss $\mathcal{L}$. The empirical effect is

$$\Delta = \mathcal{L}(R_0) - \mathcal{L}(R_\phi).$$

The claim should be accompanied by uncertainty estimates, repeated datasets or held-out groups, and an explicit accounting of model-selection trials. A larger number of tunable phase parameters can otherwise produce an apparent gain without genuine generalization.

26.5 Topological machine learning and differentiability

A persistence diagram is a combinatorial object. A neural-network pipeline using it must state whether the topological stage is fixed preprocessing, a differentiable surrogate, a smoothed approximation, or a subgradient-based optimization. The notation $\mathcal{D}_\phi(X)$ does not by itself define a differentiable map.

26.6 Subject and time leakage

For repeated time-series windows from a common subject, random window splitting can leak subject-specific structure into both train and test data. Biomedical experiments should use subject-independent splits when the target claim is generalization to unseen people. Financial experiments require temporal separation, and forward-looking labels may require purging and embargo. These safeguards should be considered part of the mathematical specification of the experiment.

Part VIII

Applications and Security

Chapter 27

Cryptographic Reconstruction

27.1 Why the historical claim is insufficient

The historical programme asserts a recursive manifold that is exponentially difficult for an attacker while being logarithmically easy for an authorized user.

No hardness proof follows from this asymmetry.

A cryptosystem requires:

$$\text{key generation}, \quad \text{encryption}, \quad \text{decryption}, \quad \text{correctness}, \quad \text{security definition}.$$

27.2 A minimal security framework

Let K be a secret seed and $\Phi(K, n)$ a phase-derived sequence.

A construction might derive a secret sequence or keystream component

$$S_n = \mathrm{KDF}(K, \Phi(K, n), n). \tag{27.1}$$

A KDF is a keying-material derivation primitive; whether the displayed use is appropriate for a keystream construction depends on the precise primitive and its security interface [19, 20]. If $\Phi(K, n)$ is deterministic given K, it is not an independent source of entropy; it is simply deterministic input to the derivation. Any security claim must therefore rely on the cryptographic security of the primitive and on the secrecy of K, not on the apparent complexity of Φ. A rigorous proposal must specify the primitive, nonce/message spaces, adversary oracle access, computational model, and security game (for example a PRF/PRG definition for a generated sequence, or IND-CPA for encryption). Only then can one state an explicit advantage bound.

The first requirement is that the public component of the phase system not make the secret directly reconstructible.

Research Program 27.1 (Phase-derived cryptography)**.** Determine whether a phase-derived construction can be reduced to a recognized computational hardness assumption. If not, compare it directly against standard cryptographic primitives and record any structural attacks discovered.

Warning

"Nonlinear", "recursive", "multidimensional", and "exponential-looking" do not constitute cryptographic hardness assumptions.

Chapter 28

Financial Regime Detection

28.1 A testable hypothesis

Let X_t denote price or mid-price and let F_t denote an information state built from order-book and trade data.

Define a phase feature map

$$\Phi_t = \Phi(F_{[t-L,t]}).$$

For a future return horizon Δ,

$$R_{t,\Delta} = \frac{X_{t+\Delta} - X_t}{X_t}.$$

The hypothesis is whether Φ_t improves prediction of

$$\mathbf{1}\{R_{t,\Delta} < -\tau\}$$

relative to baseline features.

28.2 Evaluation protocol

A valid study should use a leakage-resistant time-ordering protocol:

$$\text{walk-forward or purged walk-forward validation}, \quad \text{no future leakage},$$
$$\text{transaction costs}, \quad \text{slippage}, \quad \text{multiple-testing control}.$$

When labels use forward windows, training observations whose label intervals overlap the test interval should be purged. An embargo may be added when feature lookbacks or serial dependence create another leakage channel. Purging and embargoing are established safeguards for overlapping-label financial time series; they address specific forms of leakage but do not solve nonstationarity, multiplicity, or model-selection bias by themselves [23]. These safeguards improve evaluation integrity; they do not imply profitability.

A statistically significant predictor can still be economically useless after costs.

Research Program 28.1 (Recursive-phase financial benchmark)**.** Compare phase-aware features with:

$$\begin{gathered}\text{order-flow imbalance}, \quad \text{spread}, \quad \text{depth}, \\ \text{realized volatility}, \quad \text{standard tree models}, \quad \text{standard sequence models}.\end{gathered}$$

No statement of certainty about flash crashes is assumed.

Chapter 29

ECG and Biomedical Signal Reconstruction

29.1 Topological hypothesis

Let $X(t)$ be an ECG window. Construct an embedding

$$\mathcal{E}(X) = \{x_1, \ldots, x_m\} \subset \mathbb{R}^d$$

using a specified delay-coordinate or feature embedding.

Compute a persistence diagram

$$\mathcal{D}(X).$$

The research hypothesis is that selected persistent structures may distinguish classes of rhythm or artifact.

29.2 Motion artifact test

Let

$$X_{\text{obs}} = X_{\text{cardiac}} + X_{\text{motion}}.$$

Compare classification performance on clean and motion-corrupted recordings. When repeated windows from the same subject are present, use subject-independent splitting. Learn embedding, filtration, preprocessing, and classifier hyperparameters from training data only; otherwise subject identity or preprocessing choices can leak into the test estimate.

The central question is not whether topology "knows" the heartbeat, but whether the representation is more robust than a stated baseline.

Research Program 29.1 (Phase-aware ECG classification)**.** Evaluate ordinary TDA, phase-aware TDA, persistence-image baselines, and conventional signal/ML baselines under controlled artifact severity. Report sensitivity, specificity, AUROC, calibration, and power consumption when implemented on hardware.

No clinical efficacy claim is made without prospective validation.

Part IX

The Deep Mathematical Boundary

Chapter 30

What the Original Bible Got Right

The historical text contains several durable intuitions.

30.1 Convergent computation from multiple regions

The useful statement is:

$$\text{separated subproblems} \to \text{concurrent local elimination} \to \text{reduced interface problem.}$$

This is mathematically ordinary enough to be rigorous and open enough to permit algorithmic novelty.

30.2 Phase as a structural variable

The useful replacement for a "recursive imaginary dimension" is

$$\phi \in \Phi$$

as an actual model parameter.

Then phase can influence:

$$\text{operator coefficients}, \quad \text{filtration scales}, \quad \text{kernel weights}, \quad \text{stochastic dynamics.}$$

This is a flexible mathematical architecture without requiring a new number system.

30.3 Topology as representation

The useful insight behind "Hardware DNA" is that a representation can focus on structural invariants rather than raw coordinates.

But every such representation is a map

$$R : \mathcal{X} \to \mathcal{Y}$$

and loses whatever information lies outside the image.

The right question is therefore task sufficiency:

What tasks remain answerable from $R(x)$?

30.4 Noise rejection as a testable property

Instead of saying “noise annihilates”, define a statistic

$$\mathcal{R}(X, N)$$

and test its behavior under controlled noise levels.

A mathematical filter is judged by an error criterion, not by its name.

Chapter 31

What Cannot Be Salvaged Without New Mathematics

31.1 Universal recursive determinism

The assertion that physical randomness is universally an illusion cannot be deduced from the constructions in this notebook.

A deterministic microscopic model, even if available, would not by itself eliminate the need for probabilistic descriptions at every observational scale.

This remains outside the present mathematics.

31.2 Lossless universal topological compression

A many-to-one invariant generally cannot reconstruct the original arbitrary datum.

For a representation

$$R : X \to Y,$$

lossless recovery for all $x \in X$ requires an injective representation, or an equivalent side channel carrying the discarded information.

Hence

$$\boxed{\text{topological invariance does not imply lossless compression.}}$$

31.3 Automatic physical law from notation

Replacing a scalar by

$$\sum_n c_n i_n$$

does not derive a new physical law.

An experimentally supported physical-law claim ordinarily requires, as appropriate to the theory, well-defined state variables and equations, consistency with relevant symmetries and conservation laws, mathematical well-posedness, and empirical validation. Not every physical law must be presented with every item in the same explicit form, but the model must make clear which structural and empirical requirements apply.

31.4 Automatic cryptographic security

Recursive depth, unusual geometry, and obscure notation are not security proofs.

The security chapter therefore remains explicitly open.

Chapter 32

A Convergent Research Programme

32.1 The dependency graph

The most coherent reconstruction now has the following dependency structure:

$$\boxed{\begin{array}{l}\text{block algebra} \to \text{front decomposition} \to \text{interface Schur complement} \\ \qquad \to \text{parallel schedule} \to \text{complexity / communication analysis} \to \text{benchmark.}\end{array}}$$

A second branch is

$$\boxed{\begin{array}{l}\text{phase parameter} \to \text{operator family} \to \text{spectral properties} \\ \qquad \to \text{filter / dynamics} \to \text{topological representation} \to \text{machine learning.}\end{array}}$$

A third is

$$\boxed{\text{physical system} \to \text{measurable state} \to \text{phase feature} \to \text{prediction problem} \to \text{benchmark.}}$$

32.2 Three levels of completion

A programme is complete at one of three levels.

1. **Analytic completion:** a theorem and proof settle the stated model.
2. **Computational completion:** a reproducible finite experiment answers a predeclared benchmark to its stated uncertainty and stopping criteria.
3. **Research completion:** a conjecture is sharpened so that a future proof or disproof has a clearly bounded target.

A chapter does not become defective because deeper questions remain. It becomes defective when its stated assumptions do not determine what question is being asked.

Chapter 33

Priority Problems for the Next Revision

33.1 Priority I: prove or disprove a genuine Pulickal speedup

Build the corrected solver first.

For a sparse 3-D problem class, determine empirically and theoretically:

$$\frac{T_{\text{baseline}}}{T_{\text{Pulickal}}}, \qquad \frac{M_{\text{baseline}}}{M_{\text{Pulickal}}}, \qquad \frac{Q_{\text{baseline}}}{Q_{\text{Pulickal}}},$$

where Q denotes communication.

The first serious milestone is not "new algorithm"; it is a reproducible comparison.

33.2 Priority II: classify phase-Hodge operators

For

$$L_\phi = \sum_j w_j(\phi) L_j,$$

derive conditions for:

$$L_\phi \succeq 0, \qquad L_\phi = L_\phi^*, \qquad \text{spectral gap preservation.}$$

Then test whether the resulting filters outperform ordinary Hodge filtering.

33.3 Priority III: construct a genuinely nontrivial phase filtration

The filtration

$$K(X, \epsilon, \phi)$$

should be made sufficiently expressive that phase carries information beyond a simple rescaling of ϵ.

For example, phase could alter edge weights rather than merely the threshold:

$$w_{ij}(\phi) = g_\phi(d(x_i, x_j)). \tag{33.1}$$

Then the topological effect is no longer a trivial scalar dilation.

33.4 Priority IV: phase-aware persistence and ML

Construct a single reproducible pipeline and compare it to established baselines. Do not optimize the benchmark and the representation simultaneously on the same test set.

33.5 Priority V: physical validation

The lightning, thermal, hardware, and biomedical programmes should each have:

$$\text{model} \to \text{sensor} \to \text{dataset} \to \text{baseline} \to \text{test} \to \text{failure analysis.}$$

This creates an explicit bridge from mathematical reconstruction to engineering.

Chapter 34

Experimental Design and Reproducibility Boundary

34.1 A common experimental template

Every empirical programme in the notebook can be written as

$$\boxed{\begin{array}{l}\text{hypothesis} \to \text{data} \to \text{baseline} \to \text{metric} \to \text{split} \to \text{test} \\ \qquad\qquad \to \text{failure analysis.}\end{array}}$$

This applies to solvers, hardware, lightning, finance, ECG, thermal optimization, and phase-aware machine learning.

34.2 Baselines and tuning fairness

The baseline must be a serious competitor appropriate to the same problem class. A comparison is invalidated if the new method receives extensive tuning while the baseline receives none. The tuning budget, feature budget, compute budget, and stopping rule should be comparable.

34.3 Uncertainty and multiple comparisons

Point estimates conceal variance. Depending on the data-generating process, report confidence intervals, bootstrap intervals, repeated-run variability, or distributions over held-out subjects/time blocks. When many candidate models are searched, the number of tried variants is part of the statistical record; the best observed score is not an unbiased estimate of the performance of the selected model.

34.4 Rare events

For lightning and other rare-event tasks, define the unit of evaluation before measuring performance. Sample-wise accuracy, event-wise recall, false-alert burden, and lead time answer different operational questions. A complete benchmark should specify the event-matching and alert-merging rules.

34.5 Financial time series

Walk-forward or another strictly time-respecting evaluation protocol is the basic temporal discipline. When labels are defined over future intervals, overlapping training labels should be purged. An embargo may be required to prevent lookback overlap or serial dependence from leaking information across a test boundary. Repeated search over model variants is a separate multiplicity issue and should not be treated as solved by purging alone.

34.6 Biomedical data

Repeated observations from one person are not independent samples. Subject-wise splitting, patient-level preprocessing, and held-out-device evaluation may be necessary depending on the claim. A model that performs well only within one recording setup has not demonstrated robust clinical generalization.

34.7 Pre-registration and locked evaluation

For high-dimensional searches, development and evaluation should be separated. Features and hyperparameters may be selected using training and validation data, but the final test set should remain locked until the analysis is frozen. When many candidate models or phase definitions are tried, the search itself becomes part of the statistical history; reporting only the best observed variant understates selection uncertainty.

For systems experiments, equality of hardware and software conditions is part of the comparison. Processor count, memory limits, numerical precision, compiler settings, power budget, and preprocessing cost should be fixed or explicitly charged. A speedup obtained by silently changing these conditions is a systems result, not an intrinsic algorithmic theorem.

34.8 Reproducibility record

A complete experiment should retain code revision, environment, preprocessing parameters, data version or hash, train/validation/test assignments, hardware, random seeds where relevant, all primary and secondary metrics, and failure cases. The archive should make it possible to reproduce a negative result as readily as a positive one.

34.9 Stopping rule

Once the predeclared comparison has answered the stated question to the chosen uncertainty level, further searches should be recorded as new hypotheses rather than silently folded into the original test. This is the empirical counterpart of the mathematical stopping rule used elsewhere in the volume.

Chapter 35

Final Synthesis

The reconstructed Pulickal programme is not one theorem. It is a family of interlocking ideas with different mathematical statuses.

The central computational idea is

$$\boxed{\text{many local fronts} \to \text{reduced interface problem.}}$$

The central operator idea is

$$\boxed{\text{phase} \to \text{parameterized operator family.}}$$

The central topological idea is

$$\boxed{\text{phase} \to \text{parameterized filtration} \to \text{persistent structure.}}$$

The central information idea is

$$\boxed{\text{representation} \to \text{task-relevant structure.}}$$

The central engineering idea is

$$\boxed{\text{mathematical feature} \to \text{measurable prediction} \to \text{benchmark.}}$$

These can coexist without being collapsed into a single fictional “master equation.”

Integrity Statement

The strongest surviving interpretation of the Pulickal architecture
The architecture is best understood as a research strategy for finding useful structure in recursive, multi-region, phase-dependent systems. Its mathematical value comes from making that structure explicit and testing what follows from it.
The word "recursive" is therefore not a license to infer new mathematics. Every recursive construction in this volume earns its status by one of three routes: a proof, a falsifiable conjecture, or a reproducible experiment.

35.1 The boundary of the current draft

The following questions remain deliberately open:

1. Does a corrected Pulickal two-front ordering outperform established direct solvers for any important matrix class?
2. Which Cubix partitions minimize separator growth and communication?
3. Which phase-dependent operator families have useful stability or passivity properties?
4. Can a phase filtration capture information not already encoded by ordinary persistence?
5. Can phase-aware persistence improve robust classification under realistic noise?
6. Can a phase-derived constitutive law be physically realized?
7. Can any phase-derived cryptographic construction satisfy a recognized security definition?

These are not hidden flaws in the draft. They are the explicitly exposed frontier between what has been reconstructed and what must be discovered.

Chapter 36

Second-Pass Verification Addendum

36.1 Purpose and scope

This addendum records the final verification of the technical points that were identified during the second-pass audit. No information from the reconstruction is removed by these changes. Corrections below either sharpen a hypothesis, repair a counting or statistical statement, or make an existing status label more precise.

36.2 Verified exact results

The following statements have been rechecked algebraically at the stated scope:

- the block Schur reduction
  $$S = E - DB^{-1}C,$$
  and the corresponding reconstruction of the outer variables;
- the separated-front special case in which the outer block is block diagonal;
- the displayed 3×3 counterexample and its residual norm;
- canonical balanced-ternary existence and uniqueness;
- positivity and self-adjointness of nonnegative weighted sums of self-adjoint positive-semidefinite operators;
- the complex-measure interference expansion;
- the Gaussian-kernel positive-semidefiniteness argument;
- the stated correlated-driver Itô formula under the declared covariation model;
- the translation identity for the simple zeta deformation;
- the ordinary persistence-filtration monotonicity and the stated reparameterization test.

36.3 Resolved technical vulnerabilities

1. The Cubix dependency count is 8^L, not 2^L, when every recursion level halves the domain in all three spatial coordinates. The conclusion $L = \mathcal{O}(\log n)$ for n-scale resolution per coordinate remains unchanged.
2. The prime-resonance normalization is now explicitly restricted to i.i.d.-type settings when the simple $\widehat{\sigma}/\sqrt{N}$ denominator is used; the general statement is calibrated by the null sampling distribution or its variance.
3. The phase-conditioned capacity definition now records the transmitter/receiver information pattern explicitly. When the transmitter does not observe the state, state-dependent input laws are not admissible in the basic formulation.
4. The phase-indexed Gaussian kernel now treats phase as part of the feature argument when it varies across samples.
5. The SDE discussion distinguishes time-varying instantaneous cross-covariation from a jointly Brownian vector with constant covariance.
6. The constitutive-memory model now states an existence condition for the history integral before invoking its transform.
7. The physical-law discussion no longer suggests that every law must display the same checklist of symmetries, conservation laws, and validation steps; it instead distinguishes model requirements from empirical support.
8. The final open statements are explicitly treated as conjecture targets until their model classes, baselines, budgets, and evaluation metrics have been instantiated.

36.4 Final status boundary

The corrections above do not turn open research programmes into established results. They make the boundary sharper: proved identities remain proved identities; definitions remain definitions; experiments remain experiments; and conjectural claims are retained as explicit targets for future proof or falsification.

Chapter 37

Third-Pass Verification Addendum

37.1 Scope of the third pass

This pass rechecks the points that remained most vulnerable after the earlier repair cycle. The rule was to preserve the underlying historical proposal and existing mathematical material, while correcting only statements whose hypotheses or scope could still be misread.

37.2 Rechecked exact identities

The following identities remain valid at their stated level:

- the block Schur identity $S = E - DB^{-1}C$ when the eliminated block B is invertible;
- the separated-front specialization when the outer block is block diagonal;
- the displayed 3×3 counterexample and residual $(0, -2/3, 0)^{\top}$;
- canonical balanced-ternary representation for all integers;
- self-adjointness and positive semidefiniteness of nonnegative weighted sums on one common inner-product space;
- the complex-measure interference expansion;
- positive semidefiniteness of the Euclidean Gaussian kernel;
- the correlated-driver Itô mixed term under the declared covariation model;
- the simple zeta translation identity and transport of its zeros and pole;
- monotonicity in the ordinary filtration parameter ϵ at fixed phase.

37.3 Third-pass scope corrections

Several statements were already mathematically sound but needed sharper scope:

1. A symbol $p_\phi(\lambda)$ describes the fixed-parameter operator; if $\phi = \phi(t)$, the coefficients are time-dependent and coefficient derivatives enter the dynamics.
2. The single-letter quantity $\sup I(X; Y \mid \Phi)$ is now explicitly a benchmark unless the relevant memoryless coding theorem identifies it with operational capacity.
3. Vanishing cross-covariation implies independence only in the stated jointly Gaussian/Brownian setting, not for arbitrary continuous martingales.
4. A positive-definite M_ϕ gives a genuine anisotropic metric; positive semidefinite M_ϕ gives only a pseudometric unless further quotienting is imposed.
5. Passivity frequency tests are convention- and hypothesis-dependent; the manuscript now states the required qualification.
6. Prime-resonance standardization requires a nondegenerate estimated null variance; otherwise resampling or an exact null distribution is the appropriate calibration.

37.4 External literature cross-check

The multiparameter-persistence discussion is consistent with current literature emphasizing that the ordinary one-parameter barcode classification does not extend to general multiparameter modules; recent work also studies alternative invariants and large-scale quotients [7, 8]. Shannon's side-information work confirms that capacity depends on the state-information model at the transmitter and receiver [3]. NIST's current KDF guidance treats key derivation as a cryptographic primitive rather than an entropy source in its own right [19, 20]. Standard passivity literature likewise requires hypotheses when connecting transfer-function properties to energy inequalities [18].

37.5 Final boundary after the third pass

No formerly open application claim has been silently promoted to a theorem. No historical construction has been deleted merely because it is unproved. The repaired manuscript now places each surviving statement in one of four roles: exact mathematics, explicit definition, research programme/conjecture target, or empirical/engineering benchmark. Remaining unknowns are therefore genuine research questions rather than hidden algebraic gaps.

Appendix A

Block-Elimination Identities

A.1 Two-by-two Schur complement

For

$$A = \begin{bmatrix} B & C \\ D & E \end{bmatrix}$$

with B invertible,

$$\begin{aligned} Bx + Cy &= f, \\ Dx + Ey &= g. \end{aligned}$$

Then

$$x = B^{-1}(f - Cy),$$

and substitution gives

$$(E - DB^{-1}C)y = g - DB^{-1}f.$$

Conversely, any solution of the reduced equation together with the reconstructed x solves the original system.

A.2 Symmetric special case

If

$$A = \begin{bmatrix} B & C \\ C^\top & E \end{bmatrix}$$

is symmetric and B is symmetric positive definite, then

$$S = E - C^\top B^{-1} C$$

is the usual symmetric Schur complement.

If the full matrix is symmetric positive definite, then so is S.

This is an important special case for elliptic PDE discretizations.

Appendix B

Operator Checks

B.1 Self-adjoint weighted sums

Let $H_j = H_j^*$ with respect to one fixed common inner product.

For real coefficients w_j,

$$H = \sum_j w_j H_j$$

satisfies

$$H^* = \sum_j w_j H_j^* = H.$$

For x,

$$\langle x, Hx \rangle = \sum_j w_j \langle x, H_j x \rangle.$$

Thus nonnegative w_j preserve positive semidefiniteness.

B.2 Complex coefficient warning

If $c_j \notin \mathbb{R}$, then generally

$$\left(\sum_j c_j H_j \right)^* = \sum_j \overline{c_j} H_j \neq \sum_j c_j H_j.$$

Therefore complex phase weighting changes the spectral problem qualitatively.

Appendix C

Topological Boundary Identity

C.1 The exact algebra

For simplicial chain groups,

$$\partial_k : C_k \to C_{k-1},$$

and the fundamental chain-complex identity is

$$\boxed{\partial_{k-1} \circ \partial_k = 0.} \tag{C.1}$$

This identity should remain exact in any complex-weighted or phase-parameterized construction.

A phase may appear in coefficients or inner products, but replacing 0 by an asymptotic expression is unnecessary.

C.2 Phase-dependent chains

Let

$$C_k^\phi = C_k \otimes \mathbb{C}$$

with a ϕ-dependent inner product. Then the boundary remains

$$\partial_k^\phi = \partial_k$$

unless a genuinely different chain map is defined.

The phase dependence can instead enter the adjoint

$$(\partial_k^\phi)^*$$

through the ϕ-dependent inner product.

This provides a clean mathematical route to phase-aware Hodge theory without breaking the chain-complex axioms.

Appendix D

Experimental Reproducibility Checklist

Any computational or physical result in a future revision should record:

1. exact problem definition;
2. exact data-generation or data-acquisition process;
3. all preprocessing;
4. train/validation/test split;
5. baseline algorithms;
6. hyperparameters;
7. hardware and software environment;
8. numerical precision;
9. random seeds when randomness is used;
10. error bars or confidence intervals;
11. failure cases;
12. code or a fully reproducible algorithmic description.

A claimed improvement should be reported both in absolute and relative terms:

$$\Delta M = M_{\text{baseline}} - M_{\text{new}}, \qquad \Delta T = T_{\text{baseline}} - T_{\text{new}},$$

and, where appropriate,

$$\%\text{ improvement} = 100\frac{M_{\text{baseline}} - M_{\text{new}}}{M_{\text{baseline}}}.$$

No isolated "best case" should be used as evidence of a general law.

Appendix E

Research Questions in Formal Form

For future work, the following statements can be copied directly into a proof or experimental notebook. The items in this section are intentionally retained as *conjecture targets*: where a matrix class, machine model, stochastic signal class, task metric, or physical noise model has not yet been instantiated, the statement is not being represented as a fully formal theorem-shaped conjecture. It becomes one after those model choices are fixed.

E.1 Pulickal elimination

Conjecture Target E.1 (Possible span improvement)**.** *There exists a nontrivial, explicitly specified class of sparse matrices and a parallel machine model for which a rigorously defined two-front elimination ordering has smaller critical-path complexity than a predeclared standard baseline ordering under a stated work-comparability criterion.*

E.2 Phase-Hodge stability

Conjecture Target E.2 (Phase-stable filtering regime)**.** *There exists a nontrivial family* L_ϕ *and a specified stochastic signal class for which a phase-dependent Hodge filter has a provable robustness advantage under a predeclared loss or error metric.*

E.3 Phase-aware persistence

Conjecture Target E.3 (Nonredundant phase information)**.** *There exist datasets* X *for which a phase-parameterized persistence representation has strictly greater predictive information for a specified task than every representation in a predeclared ordinary one-parameter representation class under the same computational budget and evaluation metric.*

E.4 Prime resonance

Conjecture Target E.4 (Prime-phase deviation)**.** *There exists an explicitly defined phase statistic R, a predeclared null model, and a specified arithmetic control set for which prime inputs exhibit a statistically reproducible deviation that remains after controlling for that declared structure.*

E.5 Hardware

Conjecture Target E.5 (Phase-redundant fault tolerance)**.** *For some specified physical noise model, bandwidth/blocklength regime, and fixed energy budget, a phase-redundant code can achieve a lower undetected-error probability than a specified binary code at equal information rate under the same decoding and reliability criterion.*

Appendix F

Closing Perspective

The first version of the Pulickal project attempted to make many different phenomena converge toward one symbolic principle. This reconstruction takes the opposite route.

It asks where convergence genuinely occurs.

For the corrected elimination algorithm, convergence means reduction to a smaller interface system.

For phase operators, convergence means a controlled spectral or dynamical regime.

For topology, convergence means persistence under a specified filtration or stability notion.

For information, convergence means reduction to task-relevant structure.

For engineering, convergence means a measurable prediction or performance result.

For conjectures, convergence means that the unknown has been reduced to a statement that another mathematician can attack directly.

Thus the strongest version of Notebook II is not

one theory explaining everything.

It is

> one disciplined architecture for turning ambitious intuition into precise mathematics and testable computation.

That architecture is incomplete by design. Its unresolved problems are named rather than hidden. Its failed claims are preserved as counterexamples rather than erased. Its useful structures are separated from the stories that originally surrounded them.

The next person who opens this manuscript should be able to choose a problem, locate its assumptions, identify its current status, and continue from there.

$$\boxed{\text{Define} \to \text{derive} \to \text{test} \to \text{compare} \to \text{prove or falsify} \to \text{continue.}}$$

End of Expanded Revision

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Research Notes

For calculations, corrections, observations, and future extensions

This second volume reconstructs the parts of the Pulickal programme that extend beyond recursive-root dynamics: computational architecture, phase-parameterized operators, topology, information, hardware, stochastic systems, and applied experiments.

The historical proposals are separated from their mathematical content and rebuilt where necessary. Multi-front elimination is expressed through exact block reductions and separator structure; recursive operator ideas become explicit spectral and stability questions; and the Cubix becomes a concrete three-dimensional domain and interface architecture.

The volume then develops phase-aware topology, persistence methods, complex amplitudes, stochastic phase models, thermal and electromagnetic formulations, lightning detection, financial and biomedical experiments, and related research programmes. Established results, conjectures, experiments, and unresolved questions are kept distinct throughout.

INDEPENDENT RESEARCH MONOGRAPH

Self-Contained & Precisely Scoped